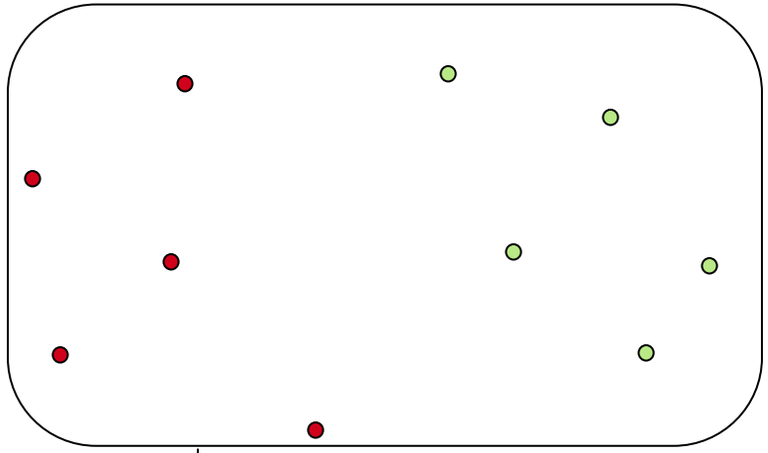
Soft-Margin SVM

Machine Learning Techniques

Karthik Thiagarajan

Hard-Margin, Linear-SVM

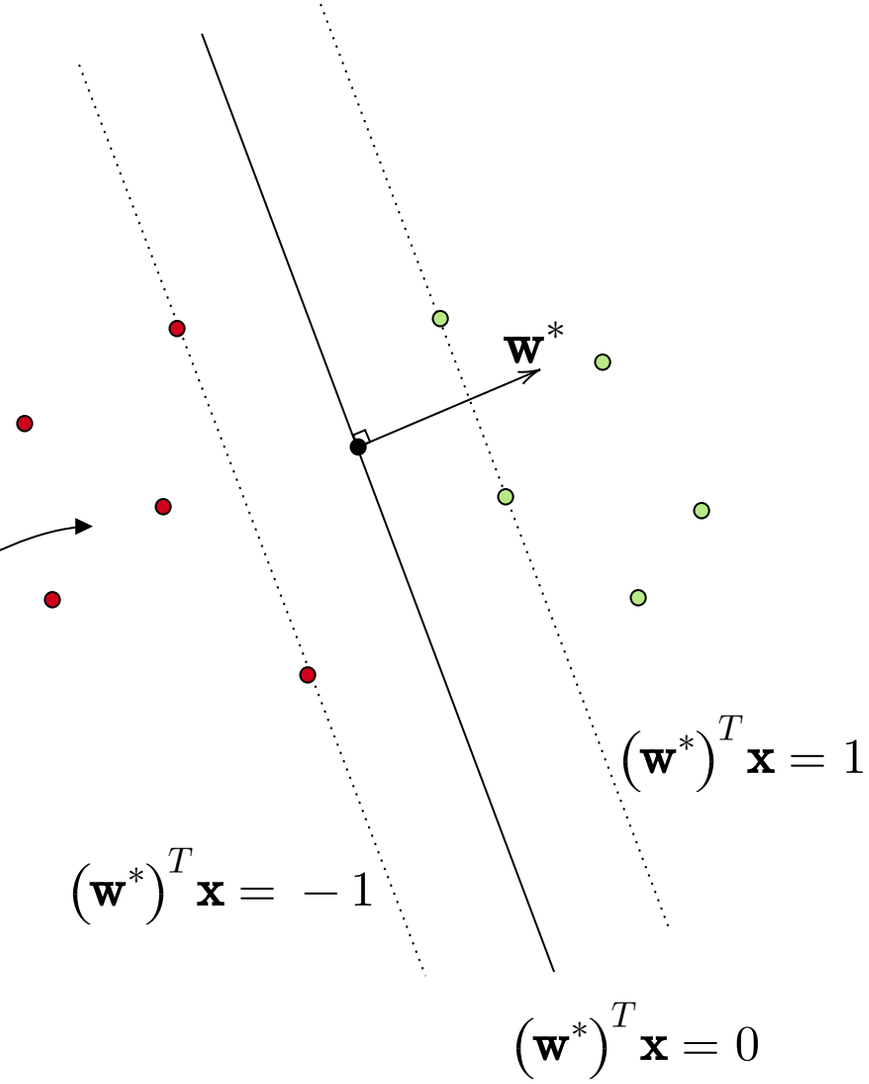


$$\widehat{y} = \begin{cases} 1, & \left(\mathbf{w}^*\right)^T \mathbf{x} \geqslant 0 \\ -1, & \left(\mathbf{w}^*\right)^T \mathbf{x} < 0 \end{cases}$$

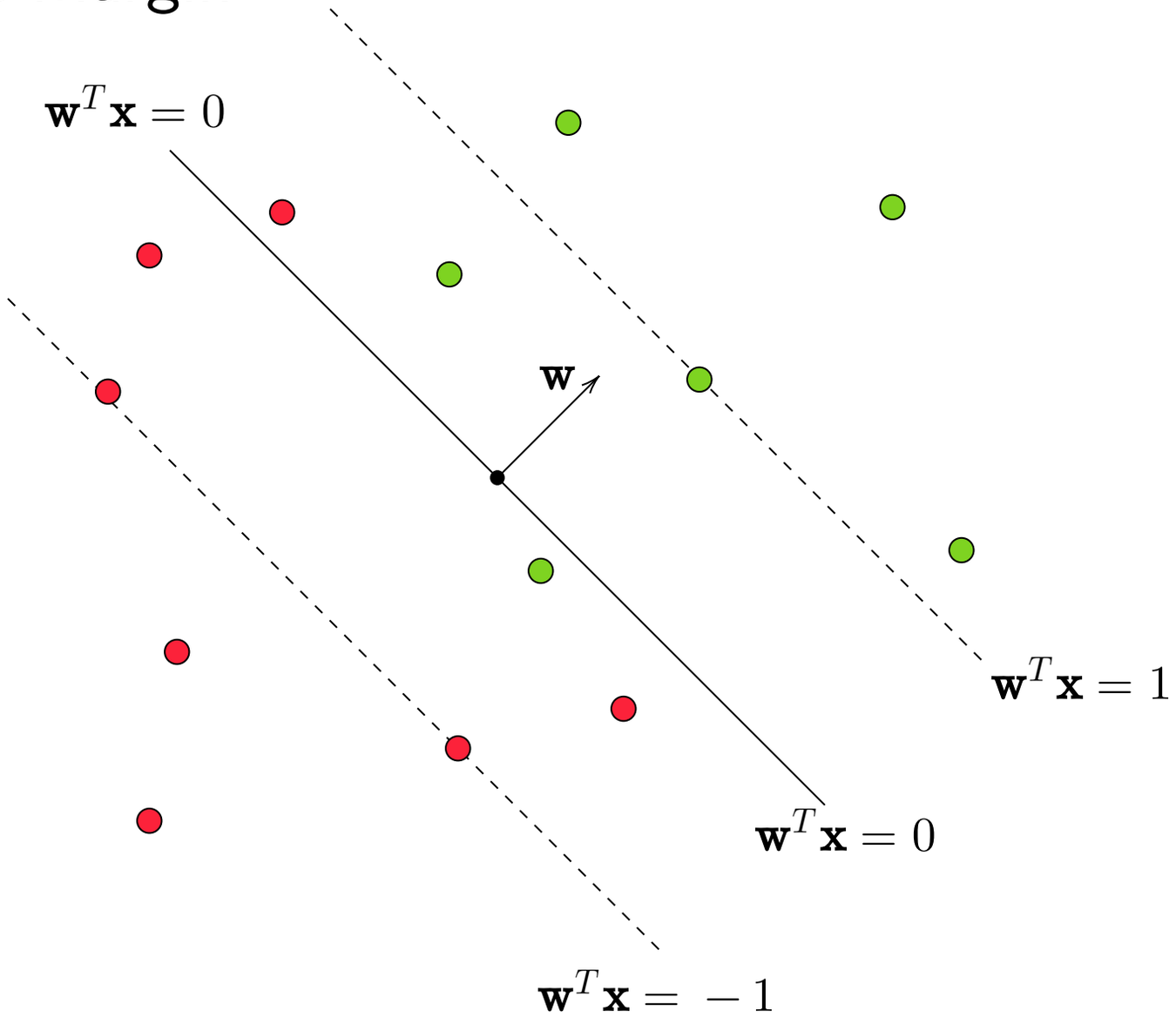
SVM Solver

$$\boldsymbol{\alpha}^* = \begin{bmatrix} \alpha_1^* \\ \vdots \\ \alpha_n^* \end{bmatrix}$$

$$\mathbf{w}^* = \sum_{i=1}^{n} \alpha_i^* \mathbf{x}_i y_i$$



Soft-Margin



$$\min_{\mathbf{w},\boldsymbol{\xi}} \quad \frac{||\mathbf{w}||^2}{2} + C \cdot \sum_{i=1}^{n} \xi_i$$

$$\left(\mathbf{w}^T\mathbf{x}_i\right) y_i + \xi_i \geqslant 1$$

$$\xi_i \geqslant 0$$

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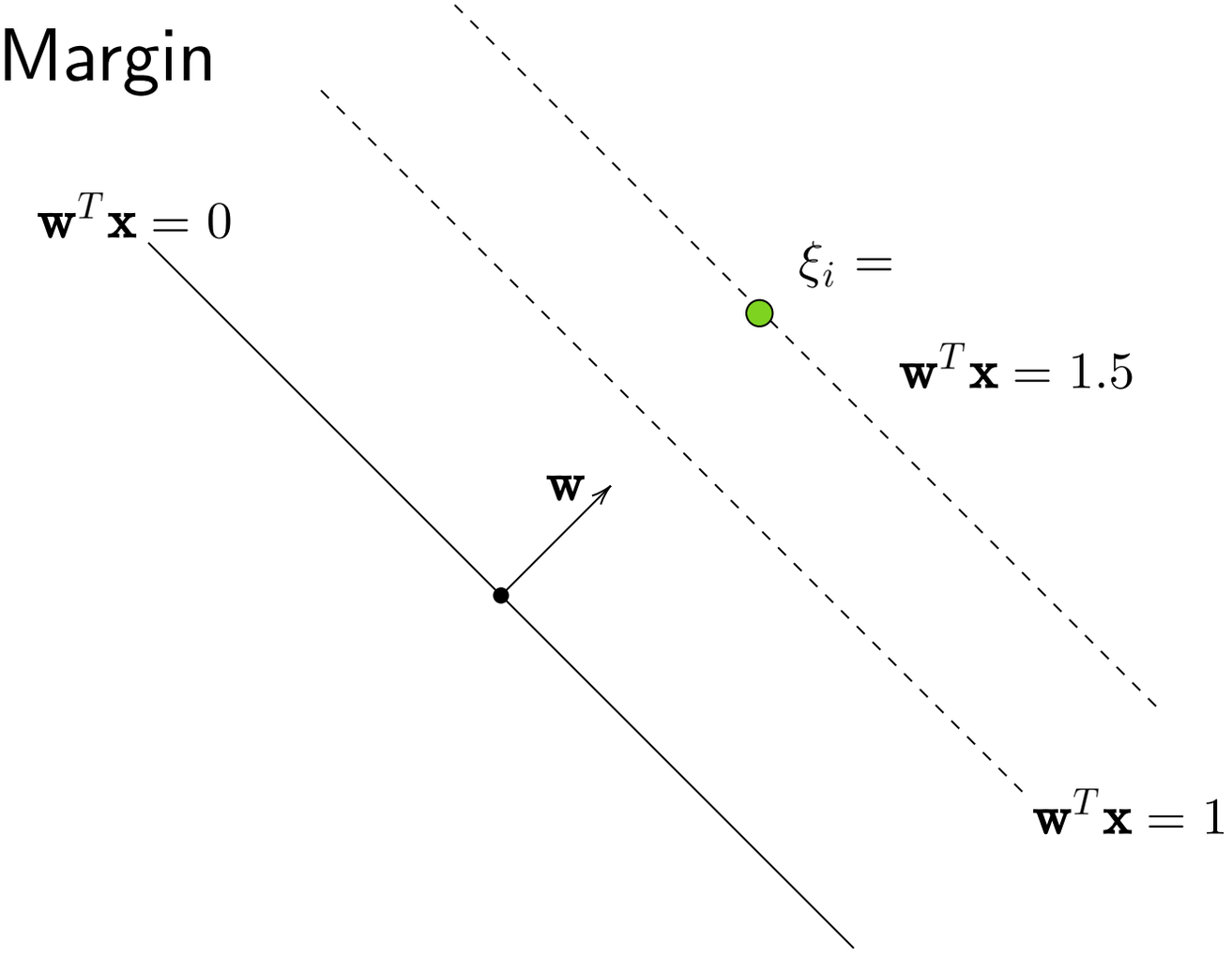

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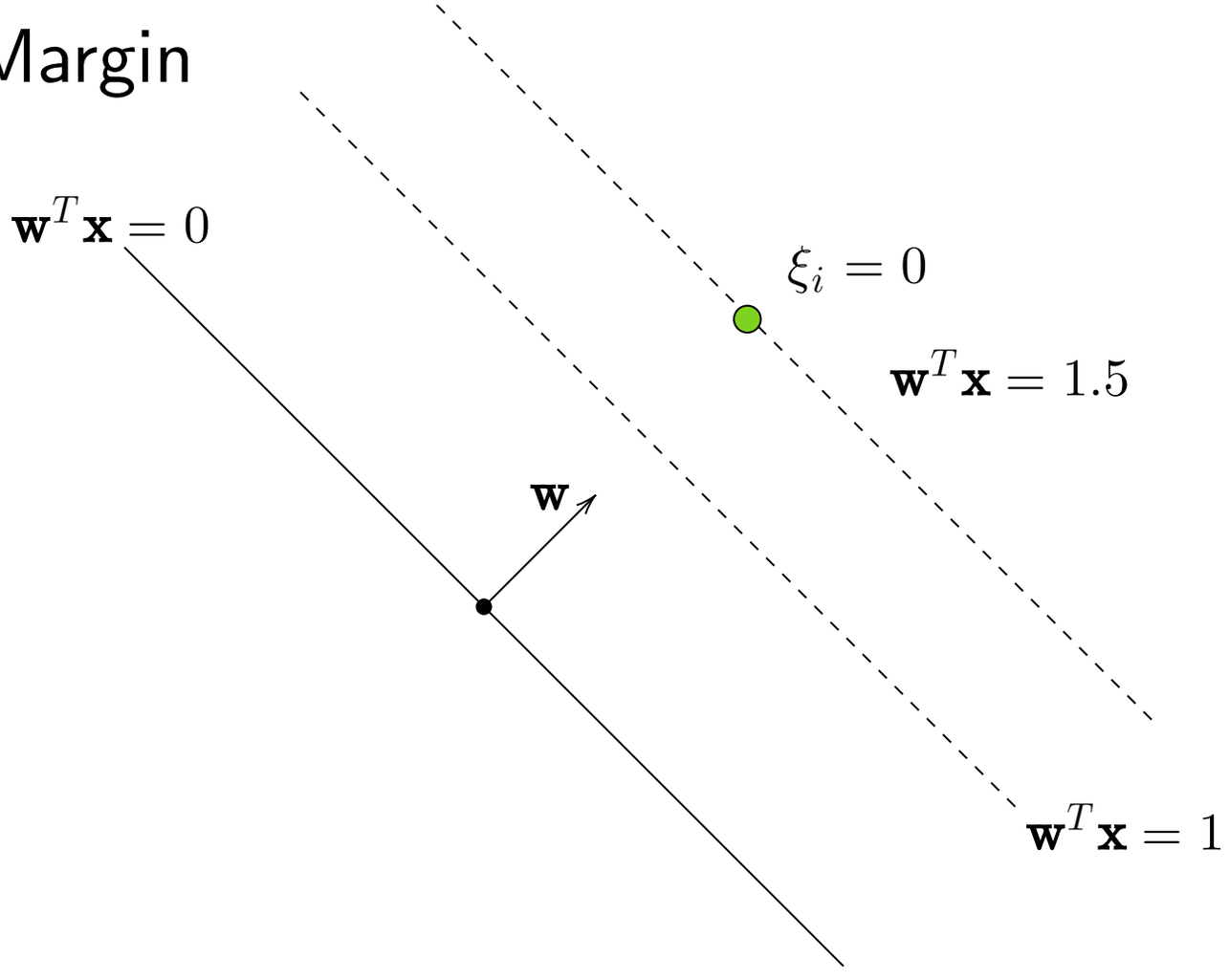

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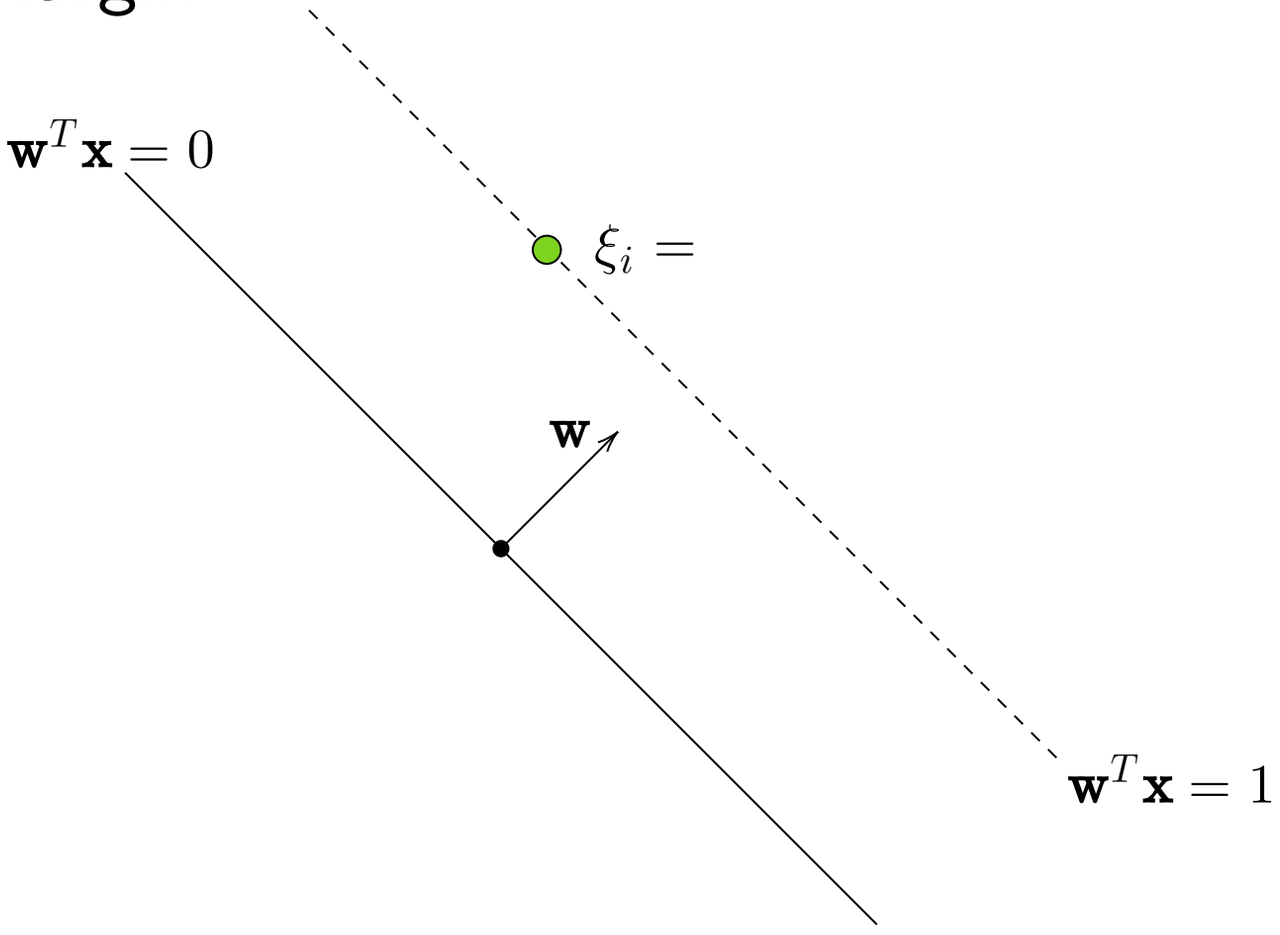

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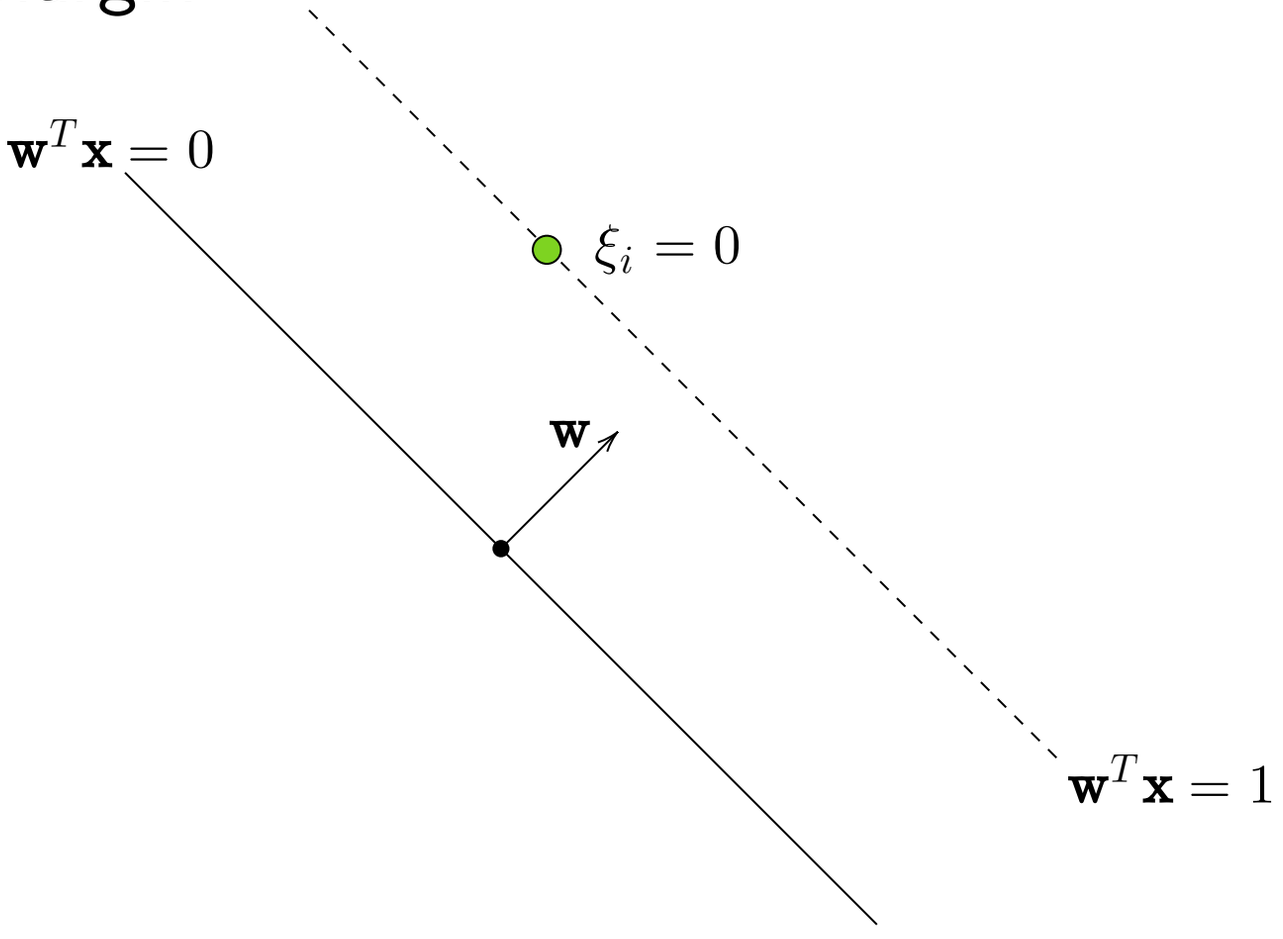

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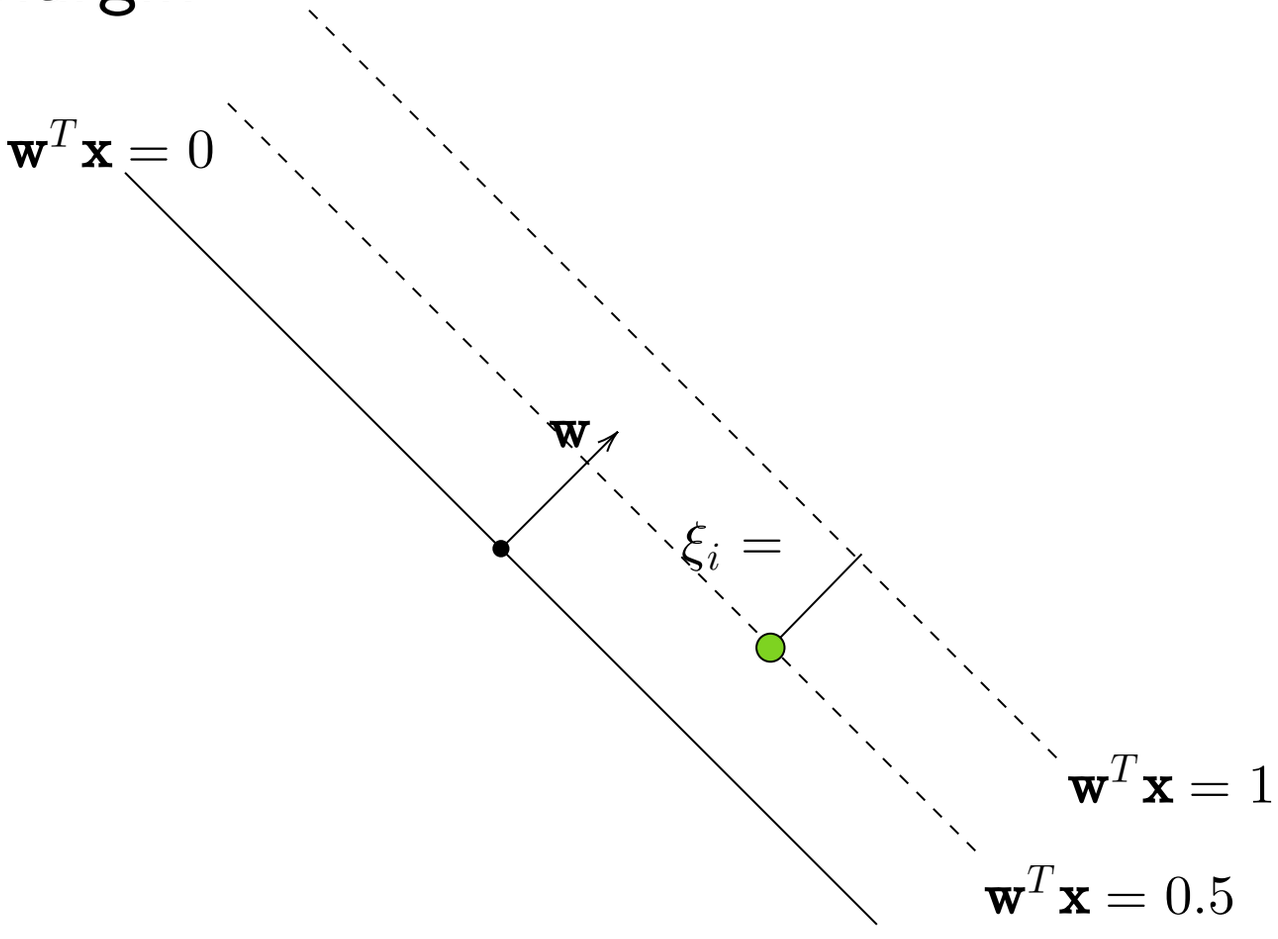

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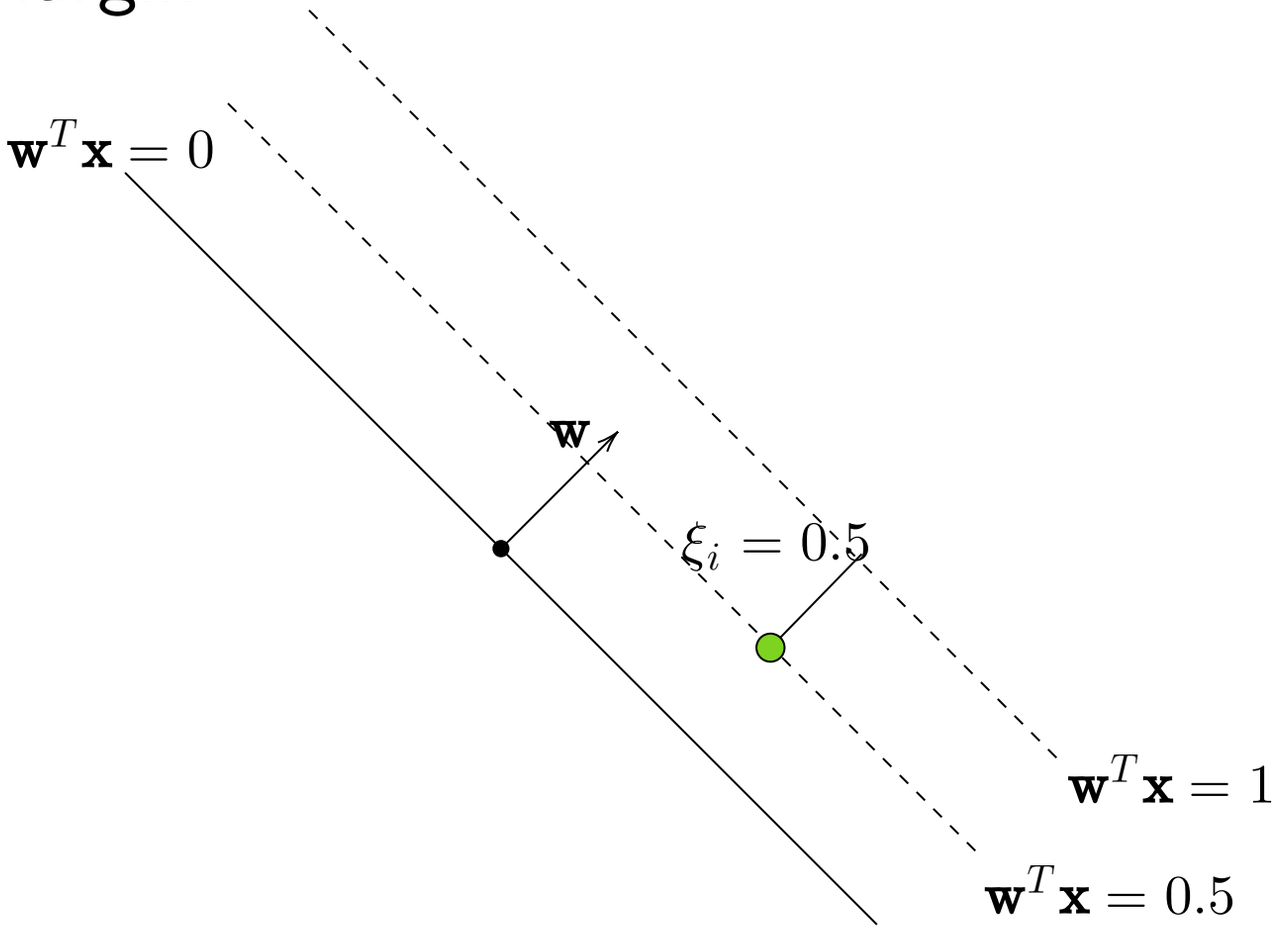

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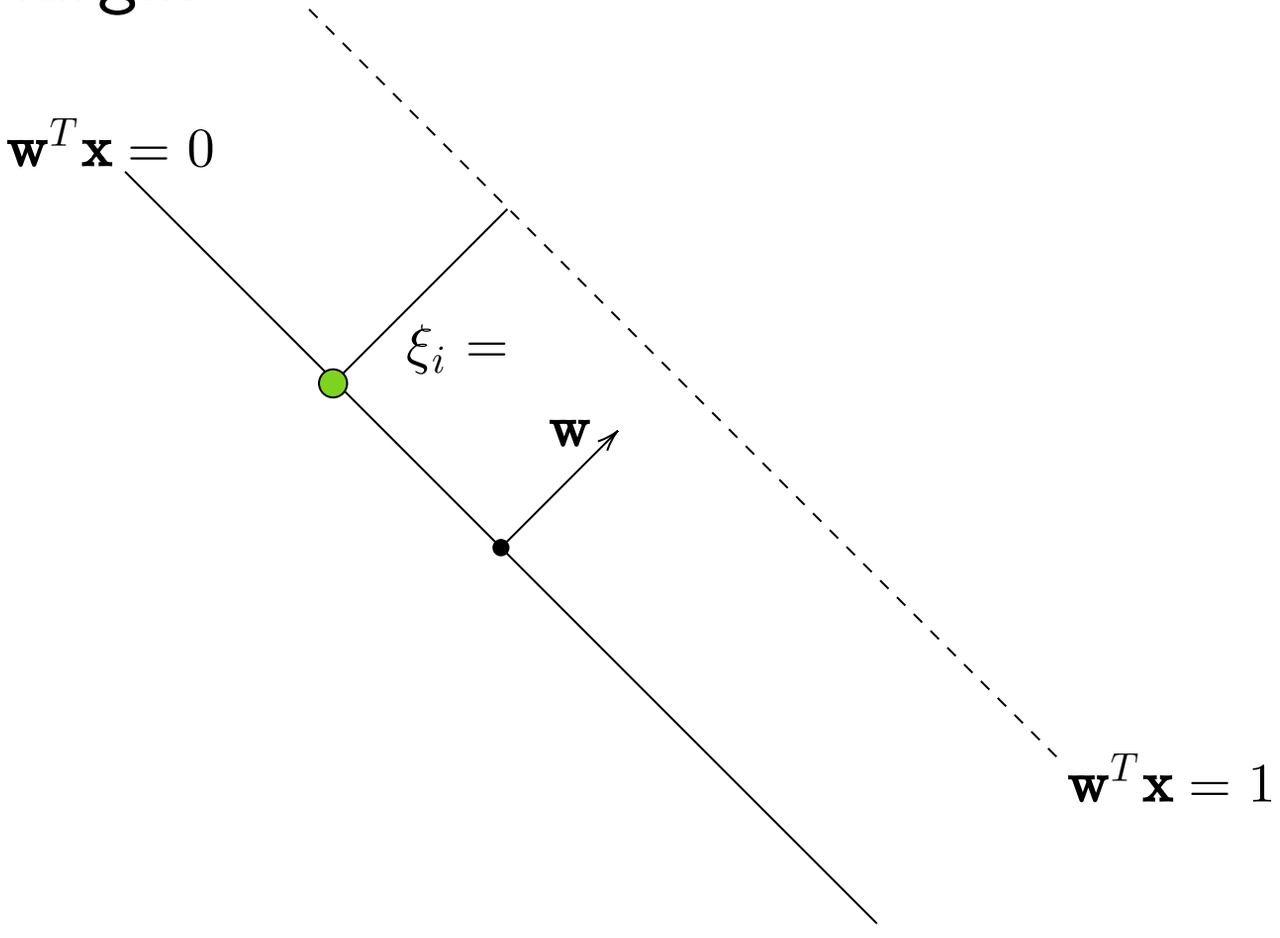

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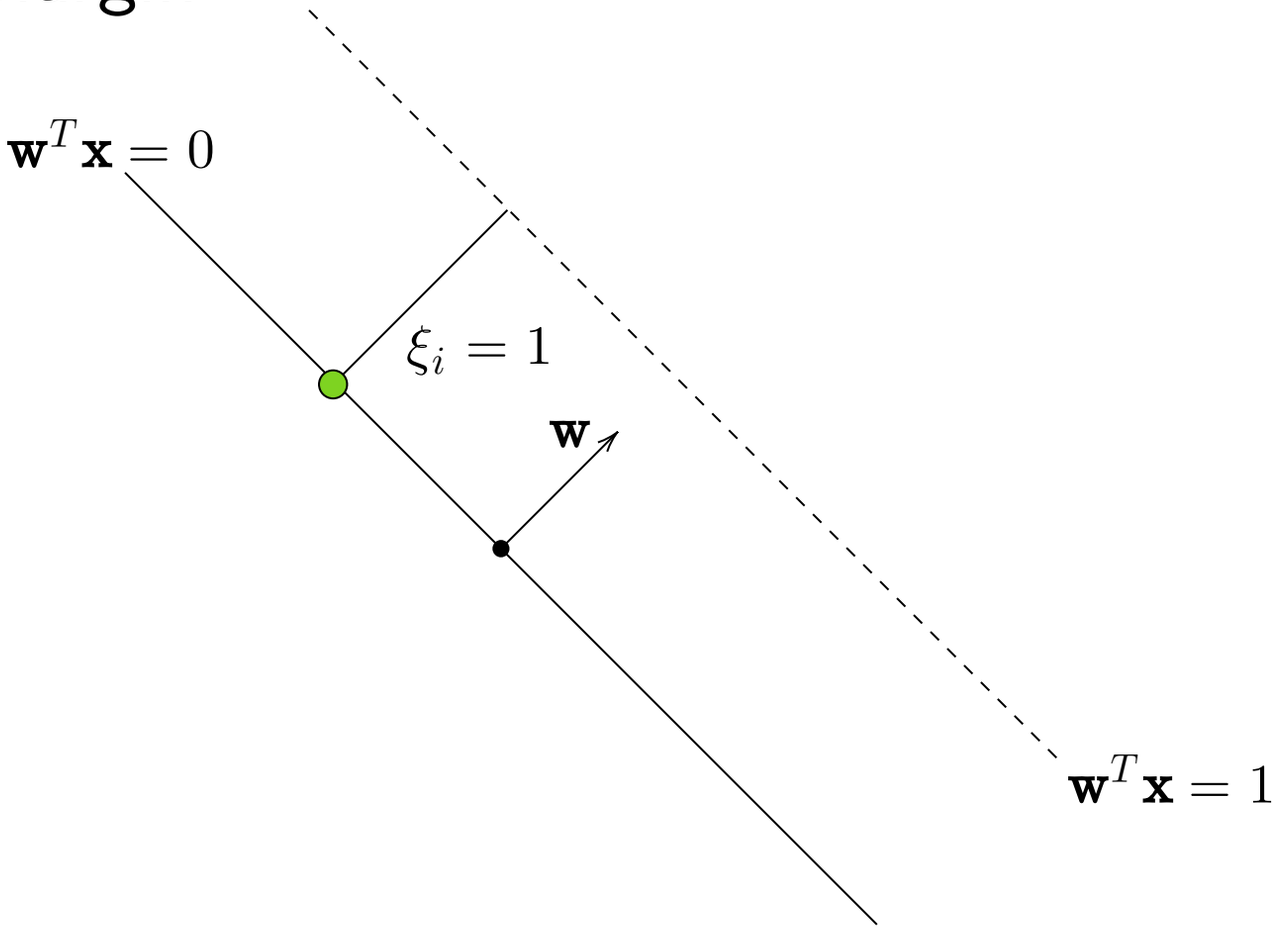

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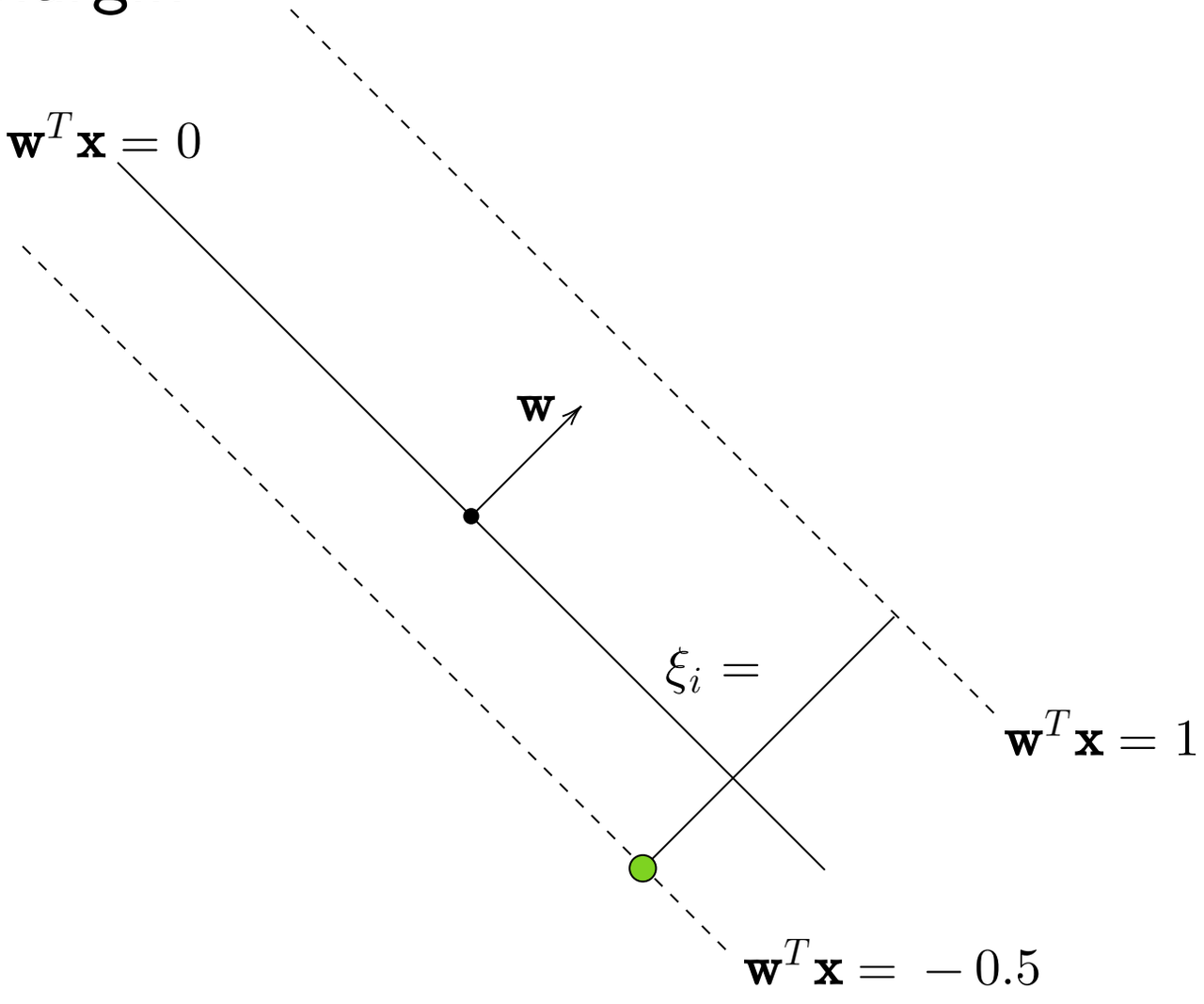

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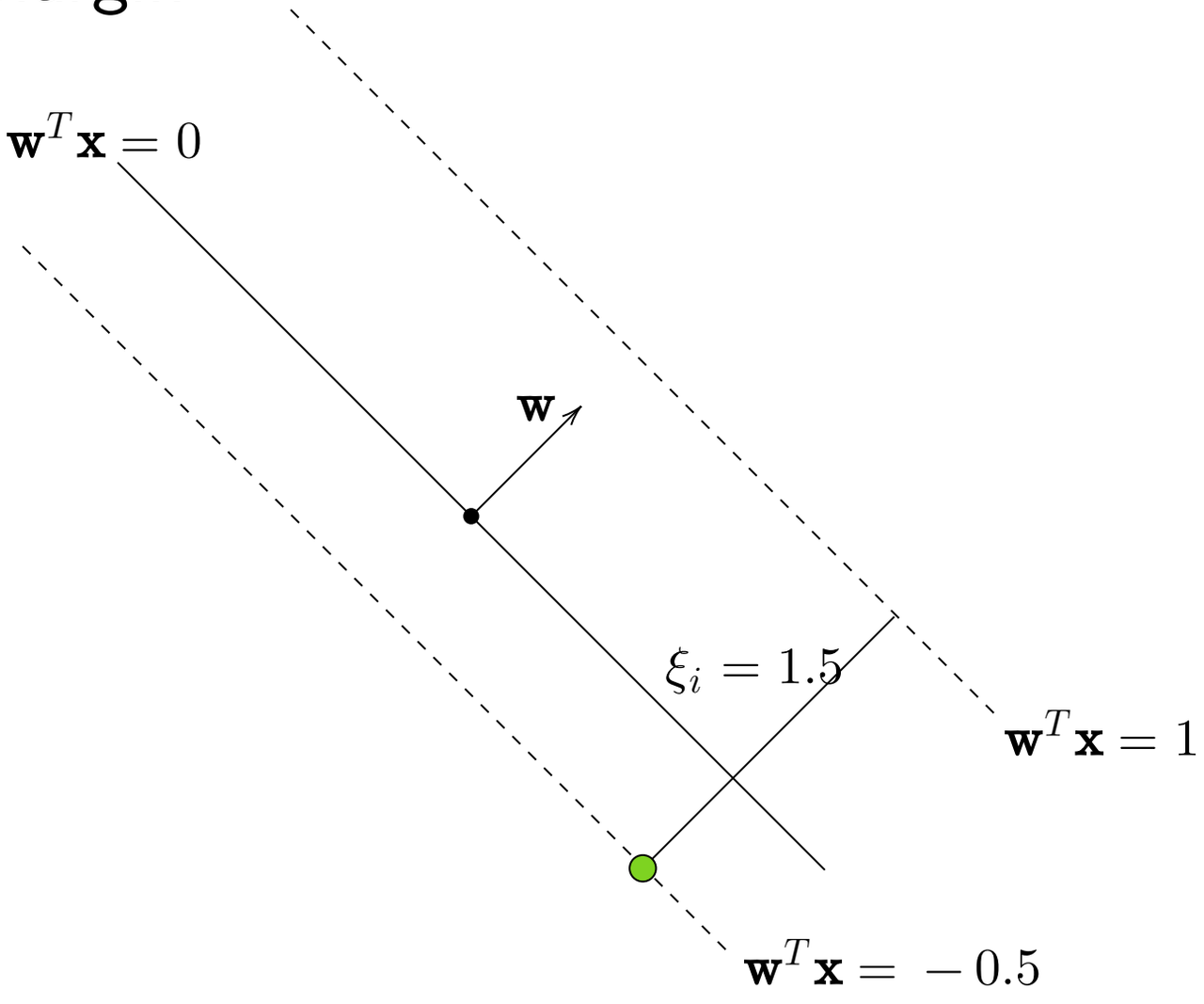

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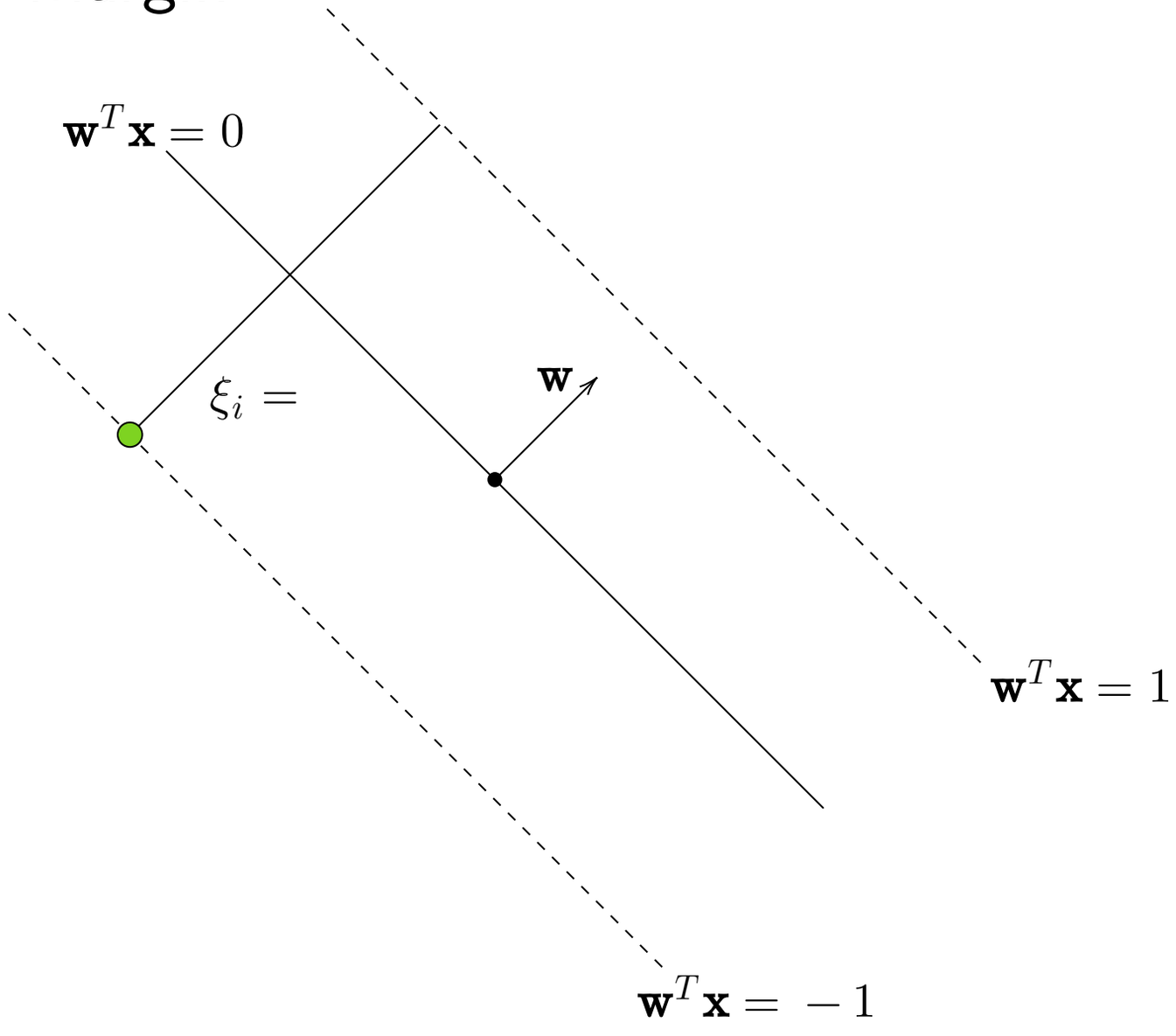

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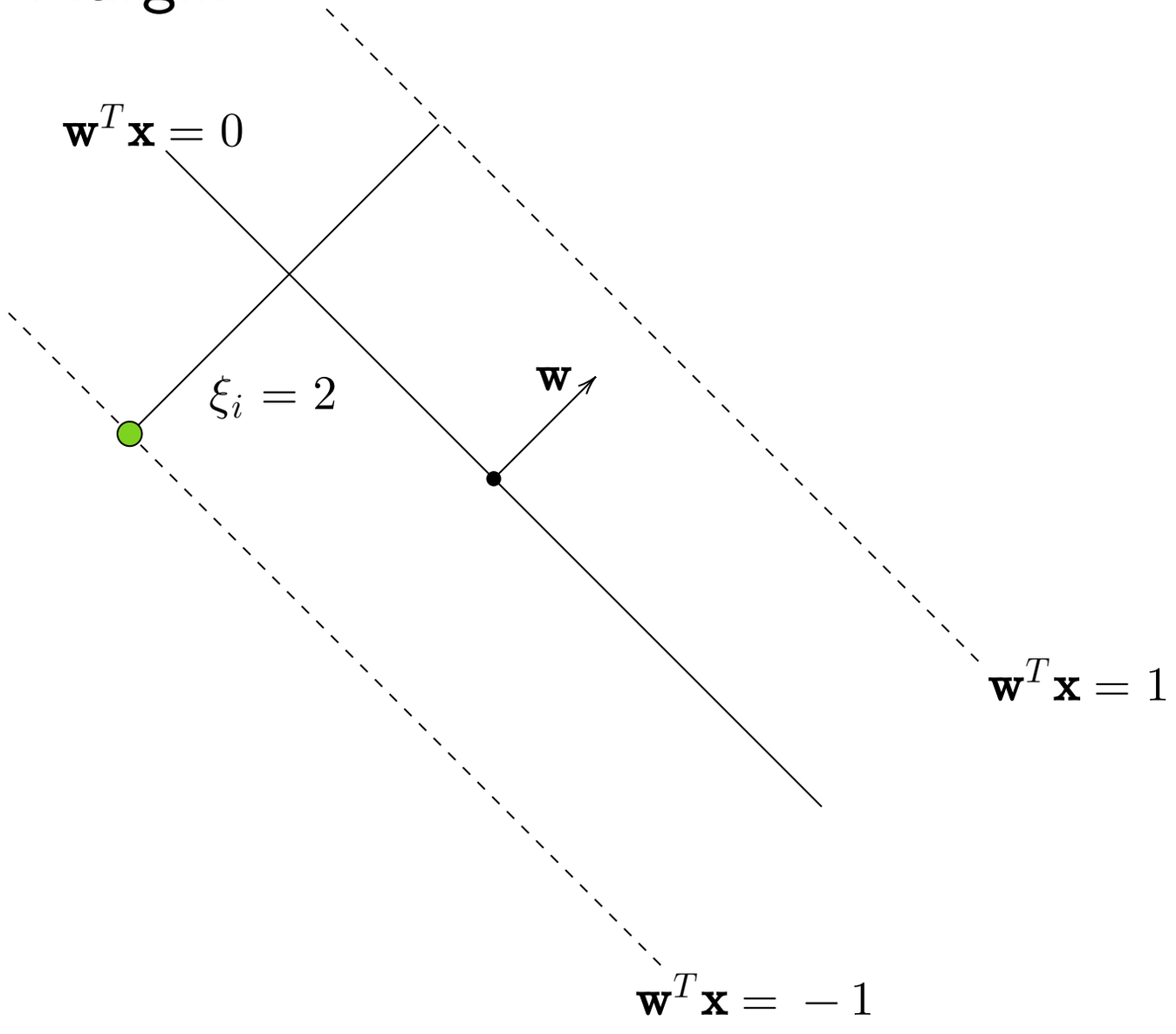

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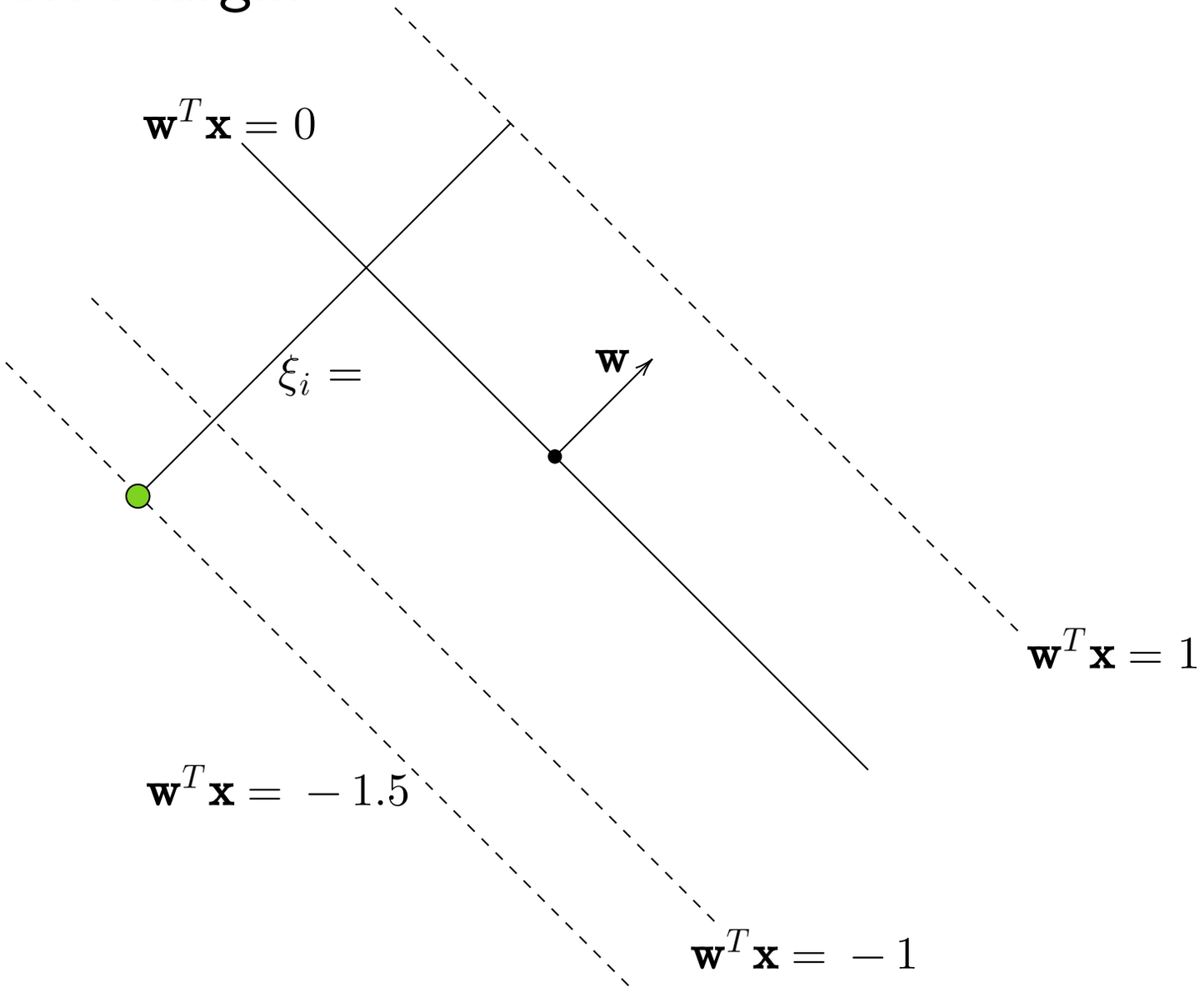

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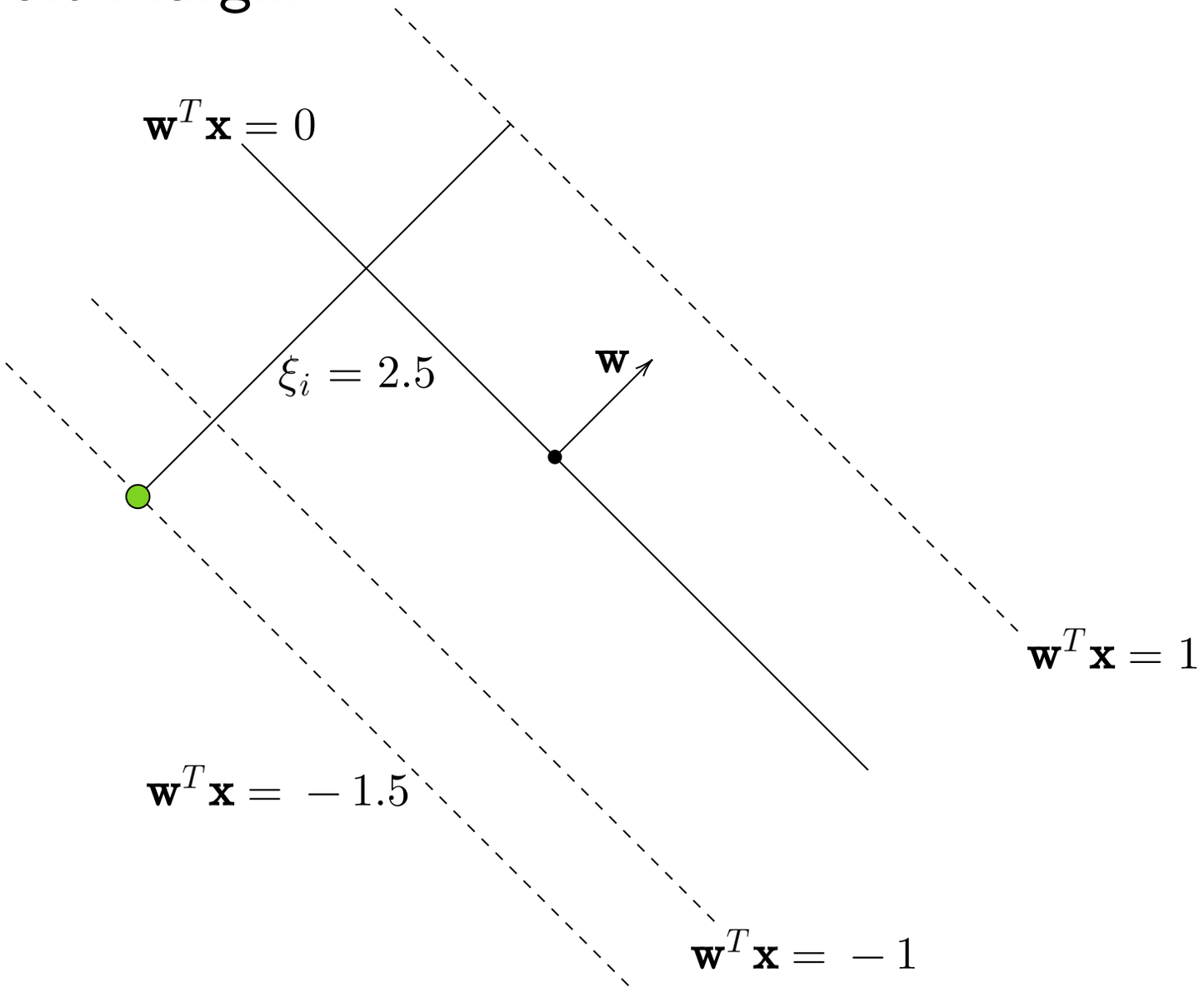

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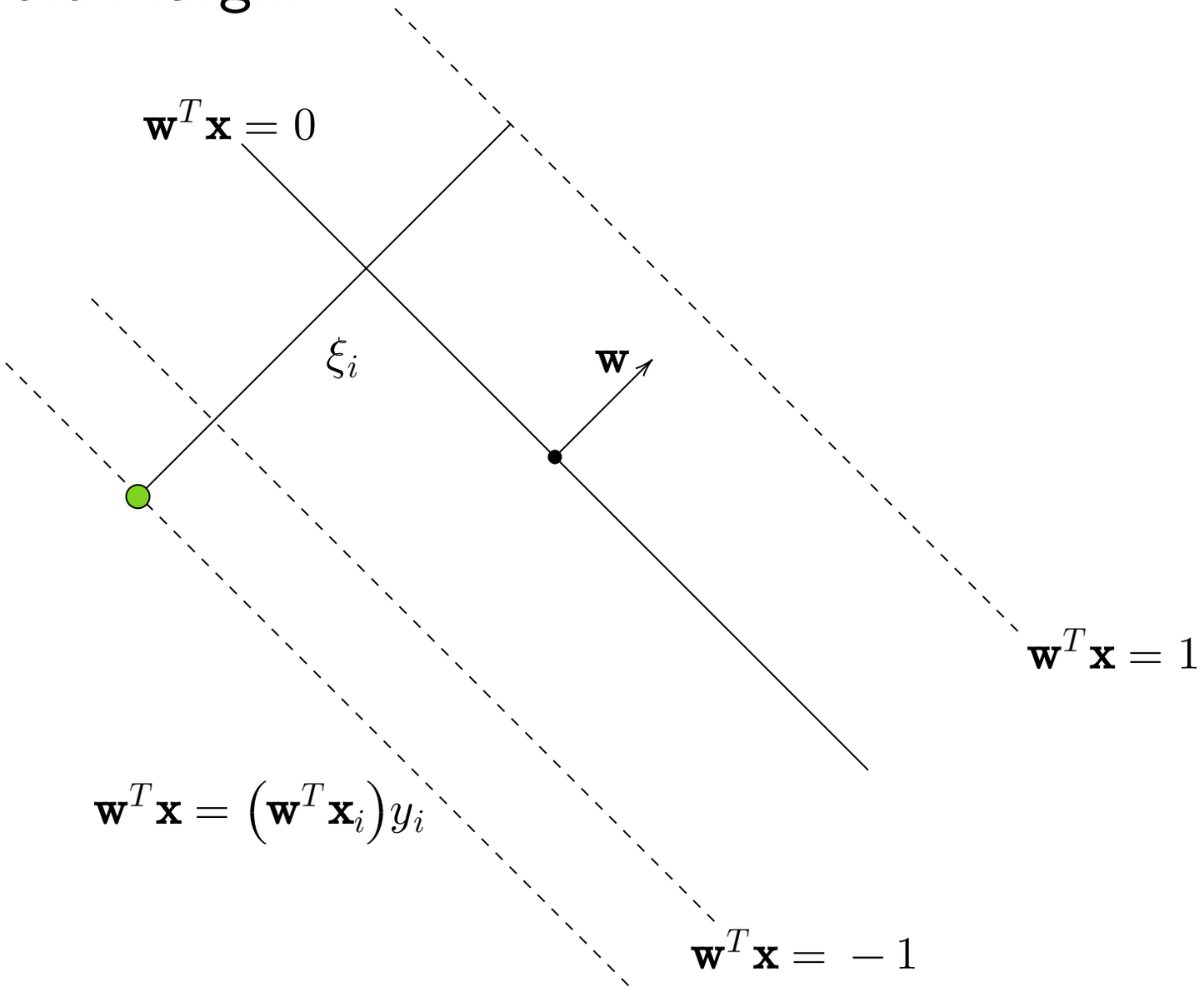

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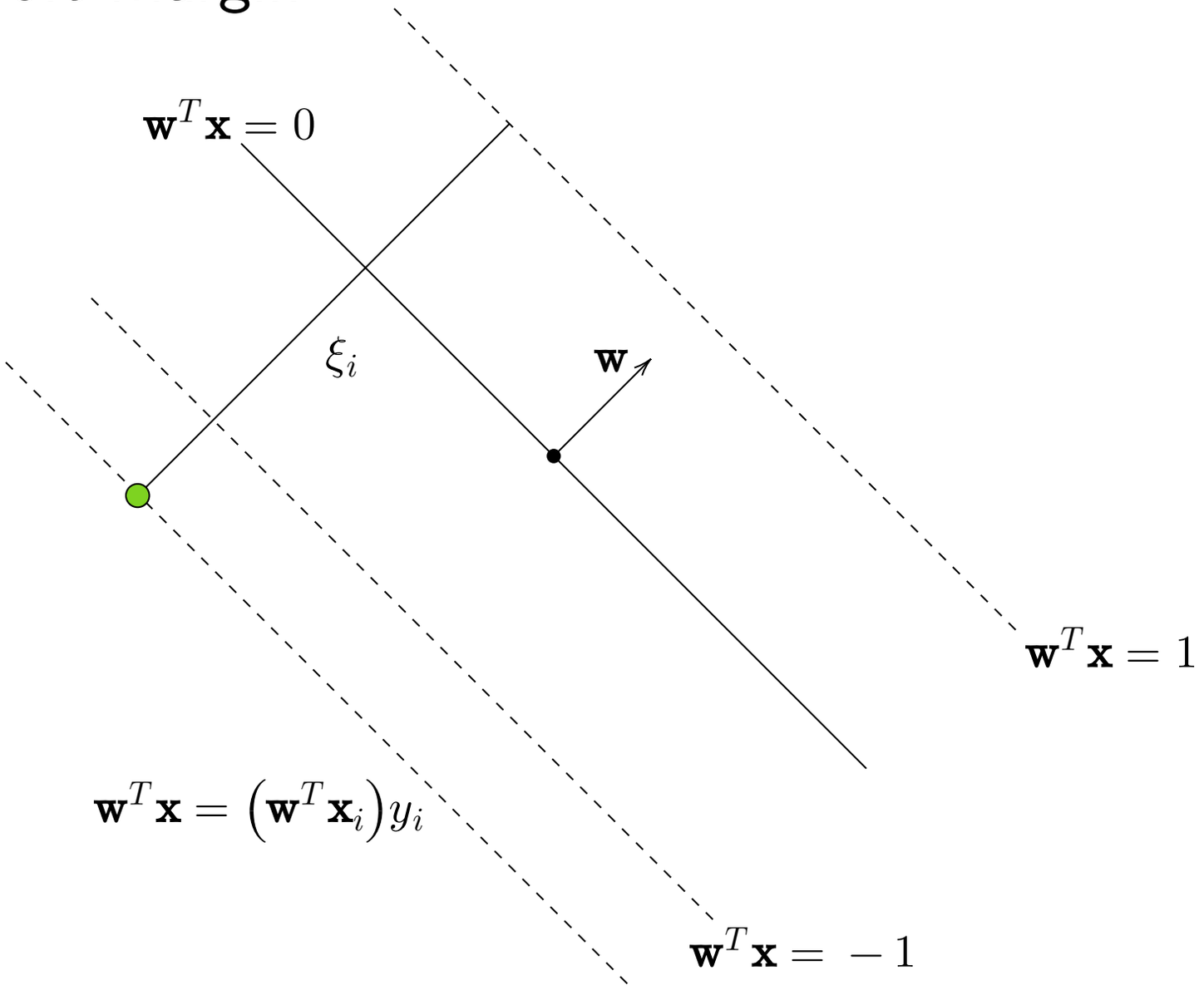


$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i + \xi_i \geqslant 1$$

$$\xi_i \geqslant 0$$

$$\boxed{\xi_i = \max\left(0, 1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right)}$$

Soft-Margin, Linear-SVM

$$\min_{\mathbf{w},\boldsymbol{\xi}} \quad \frac{||\mathbf{w}||^2}{2} + C \cdot \sum_{i=1}^{n} \xi_i$$

sub. to

$$\left(\mathbf{w}^T \mathbf{x}_i\right) y_i + \xi_i \geqslant 1, \quad 1 \leqslant i \leqslant n$$

$$\xi_i \geqslant 0, \quad 1 \leqslant i \leqslant n$$

Soft-Margin, Linear-SVM: Hinge-loss formulation

$$\min_{\mathbf{w},\boldsymbol{\xi}} \quad \frac{||\mathbf{w}||^2}{2} + C \cdot \sum_{i=1}^{n} \xi_i$$

sub. to

$$\left(\mathbf{w}^T \mathbf{x}_i\right) y_i + \xi_i \geqslant 1, \quad 1 \leqslant i \leqslant n$$

$$\xi_i \geqslant 0, \quad 1 \leqslant i \leqslant n$$

$$\equiv \qquad \min_{\mathbf{w}} \quad \frac{||\mathbf{w}||^2}{2} + C \cdot \sum_{i=1}^{n} \max\Big(0,\ 1 - \left(\mathbf{w}^T \mathbf{x}_i\right) y_i\Big)$$

Soft-Margin, Linear-SVM: Hinge-loss formulation

$$\min_{\mathbf{w},\boldsymbol{\xi}} \quad \frac{||\mathbf{w}||^2}{2} + C \cdot \sum_{i=1}^{n} \xi_i$$

sub. to

$$\left(\mathbf{w}^T \mathbf{x}_i\right) y_i + \xi_i \geqslant 1, \quad 1 \leqslant i \leqslant n$$

$$\xi_i \geqslant 0, \quad 1 \leqslant i \leqslant n$$

$$\equiv$$

Model

Data

$$\min_{\mathbf{w}} \quad \frac{||\mathbf{w}||^2}{2} + C \cdot \sum_{i=1}^{n} \max\Big(0,\ 1 - \left(\mathbf{w}^T \mathbf{x}_i\right) y_i\Big)$$

Regularization

Hinge Loss

Soft-Margin, SVM: Hinge-loss formulation

$$\min_{\mathbf{w}} \quad \underbrace{\frac{||\mathbf{w}||^2}{2}}_{\substack{(1) \\ \text{regularization}}} + C \cdot \underbrace{\sum_{i=1}^{n} \max\Big(0,\ 1 - \big(\mathbf{w}^T \mathbf{x}_i\big) y_i\Big)}_{\substack{(2) \\ \text{hinge-loss}}}$$

$\frac{||\mathbf{w}||^2}{2}$ controls the width of the margin

Smaller the value of $||\mathbf{w}||$, wider the margin

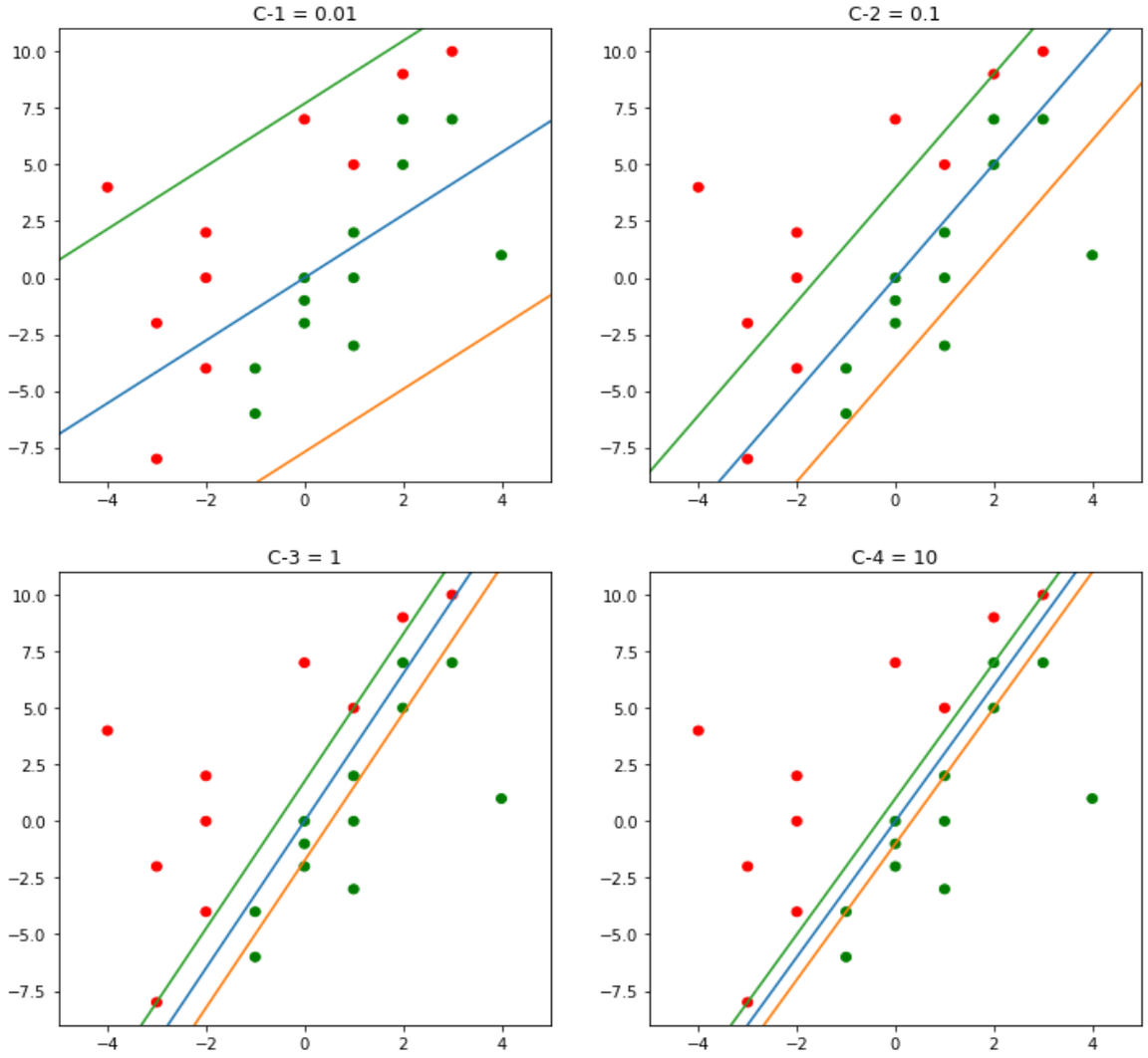
$\sum_{i=1}^{n} \max\Big(0,\ 1 - \big(\mathbf{w}^T \mathbf{x}_i\big) y_i\Big)$ is the hinge-loss.

Wider the margin, more the margin violations, greater the loss.

Soft-Margin, SVM: Hinge-loss formulation

$$\min_{\mathbf{w}} \quad \underbrace{\frac{||\mathbf{w}||^2}{2}}_{(1)} + C \cdot \underbrace{\sum_{i=1}^{n} \max\Big(0,\ 1 - \big(\mathbf{w}^T \mathbf{x}_i\big) y_i\Big)}_{(2)}$$

- (1) and (2) work in opposite directions
- If $||\mathbf{w}||$ decreases, the margin becomes wider, which increases the hinge-loss.
- C controls the tradeoff between (1) and (2):
  - If C is small, we are fine with a wide margin.
  - If C is large, we prefer a narrow margin.
  - If $C \to \infty$, we do not tolerate any violations (hard-margin)



Soft-Margin, Optimization

$$\min_{\mathbf{w},\boldsymbol{\xi}} \quad \frac{||\mathbf{w}||^2}{2} + C \cdot \sum_{i=1}^{n} \xi_i$$

$$\left(\mathbf{w}^T \mathbf{x}_i\right) y_i + \xi_i \geqslant 1$$

$$\xi_i \geqslant 0$$

Primal

$$\min_{\mathbf{w},\boldsymbol{\xi}} \quad \max_{\boldsymbol{\alpha},\boldsymbol{\beta}\geqslant\mathbf{0}} \quad \frac{||\mathbf{w}||^2}{2} + C \cdot \sum_{i=1}^{n} \xi_i + \sum_{i=1}^{n} \alpha_i \Big[1 - \left(\mathbf{w}^T \mathbf{x}_i\right) y_i - \xi_i\Big] - \sum_{i=1}^{n} \beta_i \xi_i$$

Dual

$$\max_{\boldsymbol{\alpha},\boldsymbol{\beta}\geqslant\mathbf{0}} \quad \min_{\mathbf{w},\boldsymbol{\xi}} \quad \frac{||\mathbf{w}||^2}{2} + C \cdot \sum_{i=1}^{n} \xi_i + \sum_{i=1}^{n} \alpha_i \Big[1 - \left(\mathbf{w}^T \mathbf{x}_i\right) y_i - \xi_i\Big] - \sum_{i=1}^{n} \beta_i \xi_i$$

Soft-Margin, Optimization

$$\max_{\boldsymbol{\alpha},\boldsymbol{\beta}\geqslant\mathbf{0}} \quad \underbrace{\min_{\mathbf{w},\boldsymbol{\xi}} \quad \frac{||\mathbf{w}||^2}{2} + C\cdot\sum_{i=1}^{n}\xi_i + \sum_{i=1}^{n}\alpha_i\Big[1-\big(\mathbf{w}^T\mathbf{x}_i\big)y_i - \xi_i\Big] - \sum_{i=1}^{n}\beta_i\xi_i}_{\mathbf{w}=\mathbf{X}\mathbf{Y}\boldsymbol{\alpha},\ \ \alpha_i+\beta_i=C}$$

$$\boxed{\max_{0\leqslant\boldsymbol{\alpha}\leqslant C\cdot\mathbf{1}} \quad \boldsymbol{\alpha}^T\mathbf{1} - \frac{1}{2}\boldsymbol{\alpha}^T\big[\mathbf{Y}^T\mathbf{X}^T\mathbf{X}\mathbf{Y}\big]\boldsymbol{\alpha}}$$

$0 \leqslant \alpha_i \leqslant C$

Box constraints

Soft-Margin, Optimization

$$\min_{\mathbf{w},\boldsymbol{\xi}} \quad \frac{||\mathbf{w}||^2}{2} + C \cdot \sum_{i=1}^{n} \xi_i$$

Primal constraints

$$\left(\mathbf{w}^{*T}\mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0$$

$$\max_{\boldsymbol{\alpha}} \quad \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2}\boldsymbol{\alpha}^T \left[\mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}\right] \boldsymbol{\alpha}$$

Dual constraints

$$0 \leqslant \alpha_i^* \leqslant C$$

$$\mathbf{w}^* = \mathbf{X}\mathbf{Y}\boldsymbol{\alpha}^*$$

$$\alpha_i^* + \beta_i^* = C$$

Complementary Slackness

$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left(\mathbf{w}^{*T}\mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$

Soft-Margin, Complementary Slackness, Primal View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

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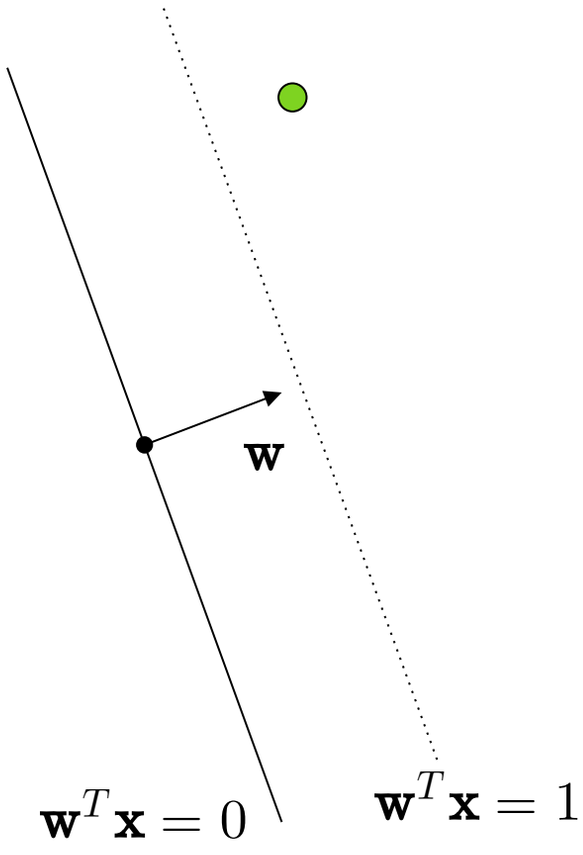

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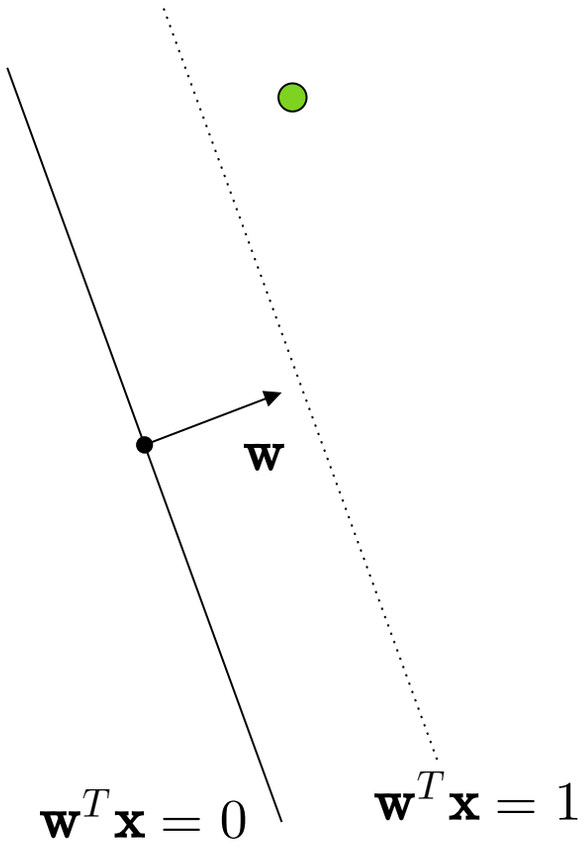

$$\boxed{\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i > 1}$$

Soft-Margin, Complementary Slackness, Primal View

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$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

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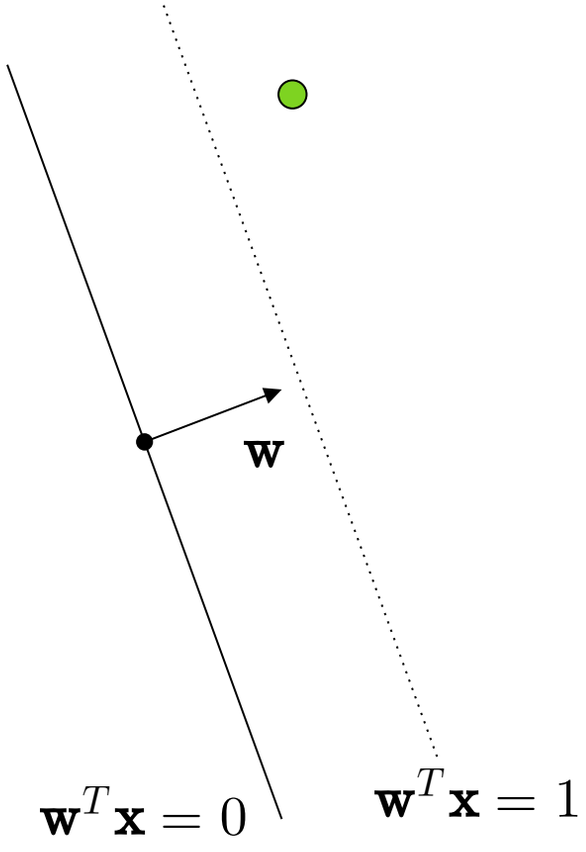

$$\boxed{\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i > 1}$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$

$$\boxed{\alpha_i^* = 0}$$

Soft-Margin, Complementary Slackness, Primal View

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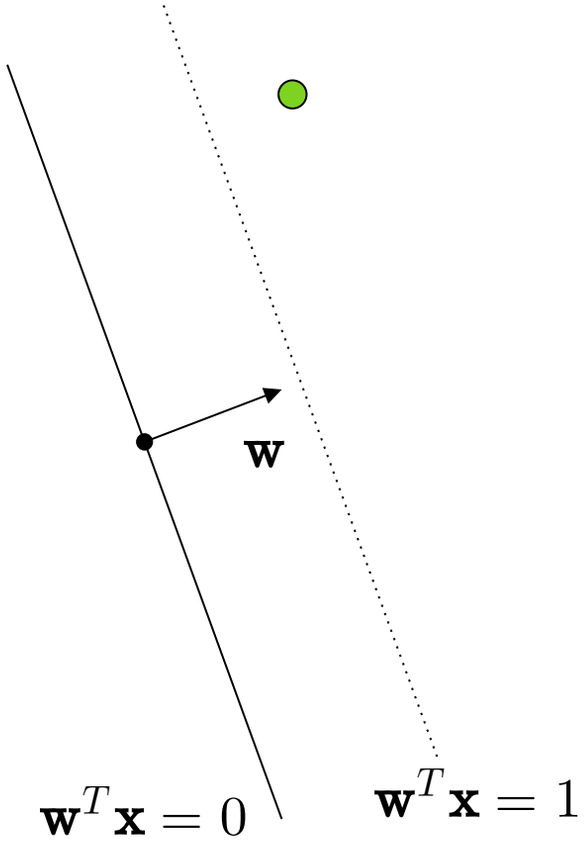

$$\boxed{\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i > 1}$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$

$$\boxed{\alpha_i^* = 0}$$

$$\alpha_i^* + \beta_i^* = C$$

$$\beta_i^* = C$$

Soft-Margin, Complementary Slackness, Primal View

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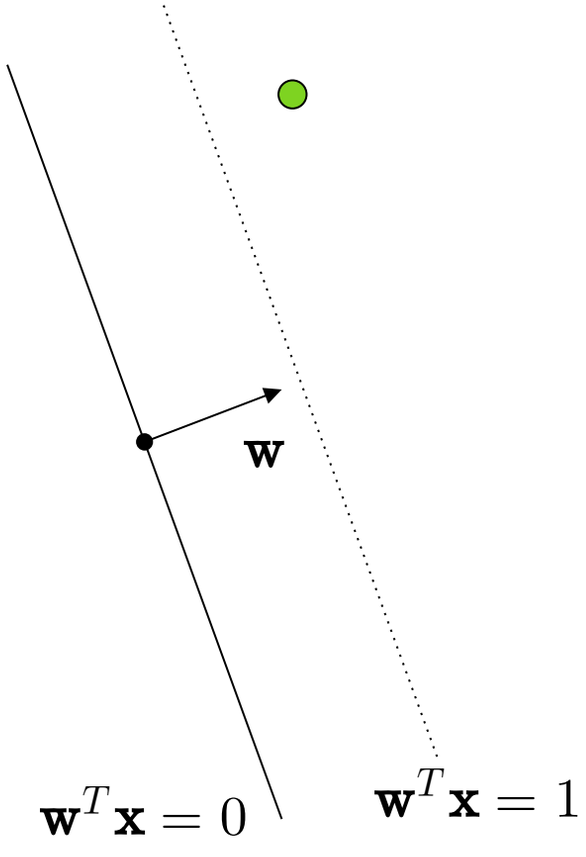

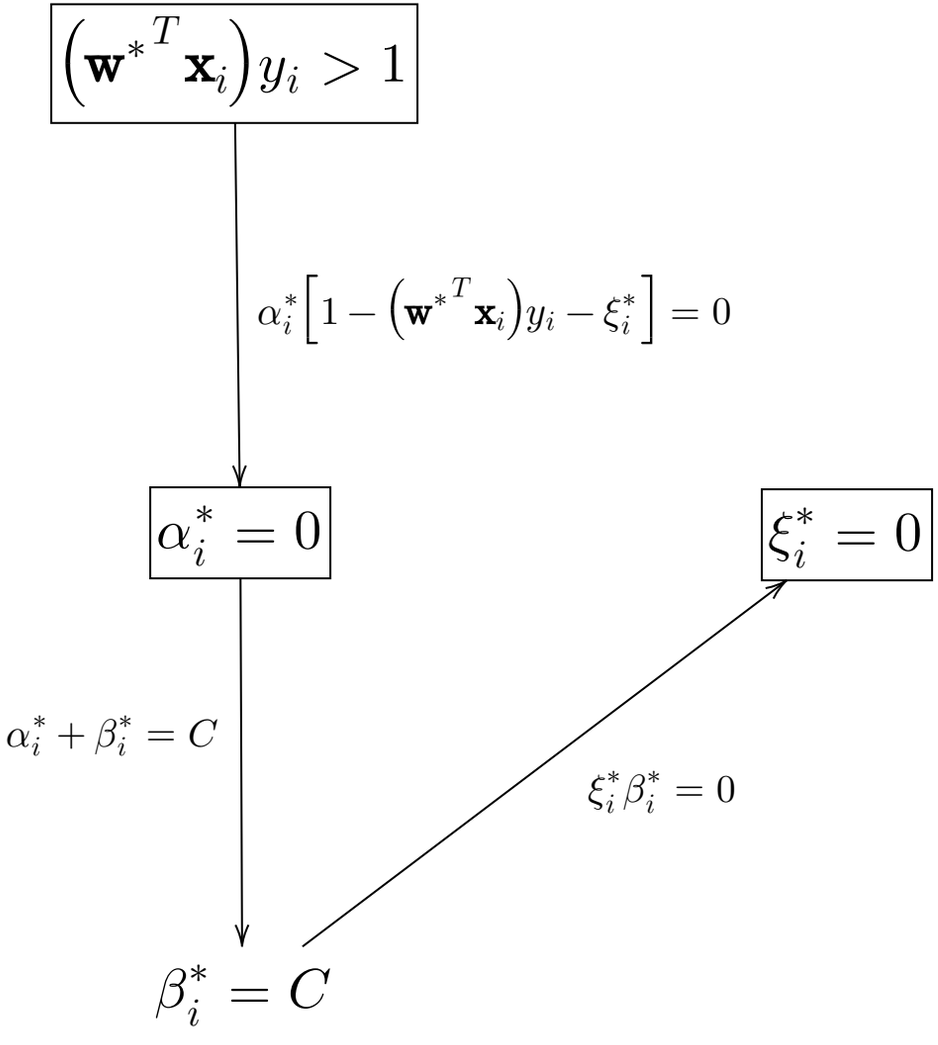

Soft-Margin, Complementary Slackness, Primal View

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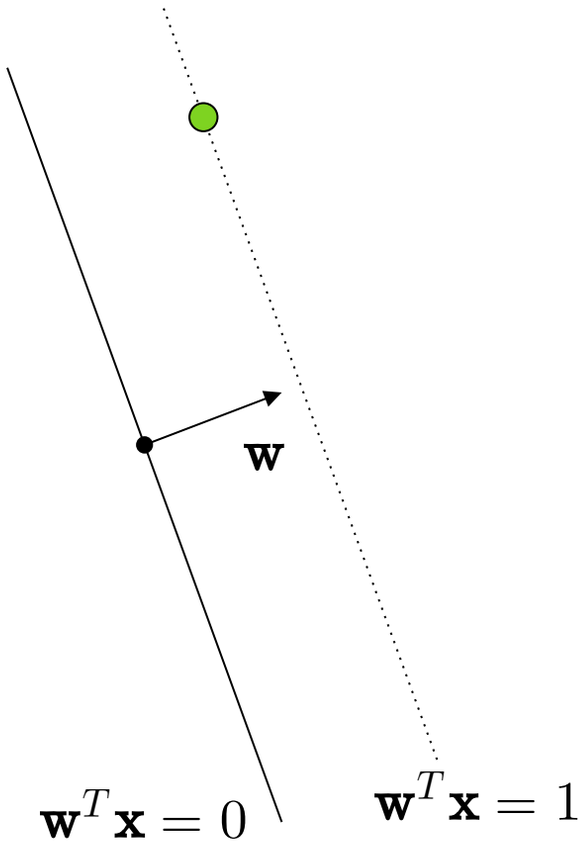

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$$\xi_i^* \beta_i^* = 0$$

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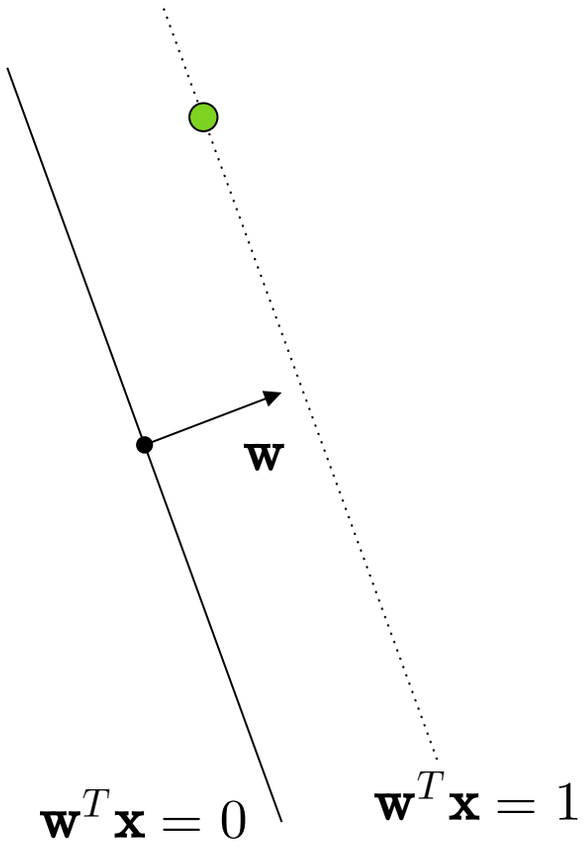


$$\boxed{\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i = 1}$$

Soft-Margin, Complementary Slackness, Primal View

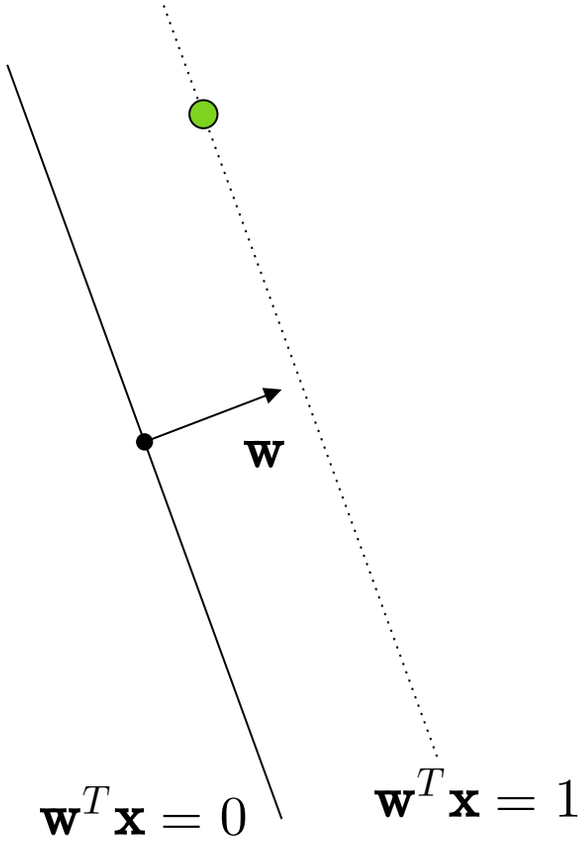
$$\left(\mathbf{w}^{*T}\mathbf{x}_i\right)y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left(\mathbf{w}^{*T}\mathbf{x}_i\right)y_i - \xi_i^*\right] = 0$$



$$\boxed{\left(\mathbf{w}^{*T}\mathbf{x}_i\right)y_i = 1}$$

$$\Big\downarrow \; 1 - \left(\mathbf{w}^{*T}\mathbf{x}_i\right)y_i - \xi_i^* \leqslant 0$$

$$\boxed{\xi_i^* = 0}$$

Soft-Margin, Complementary Slackness, Primal View

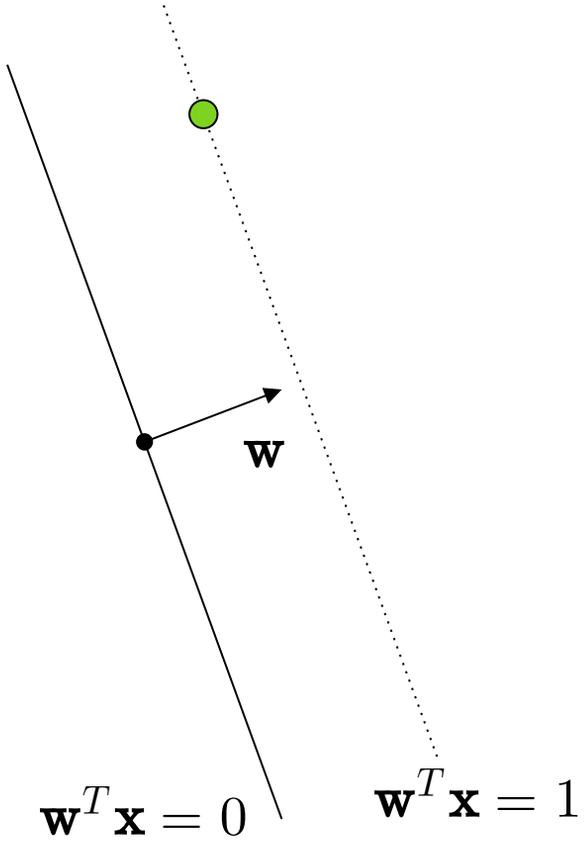
$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

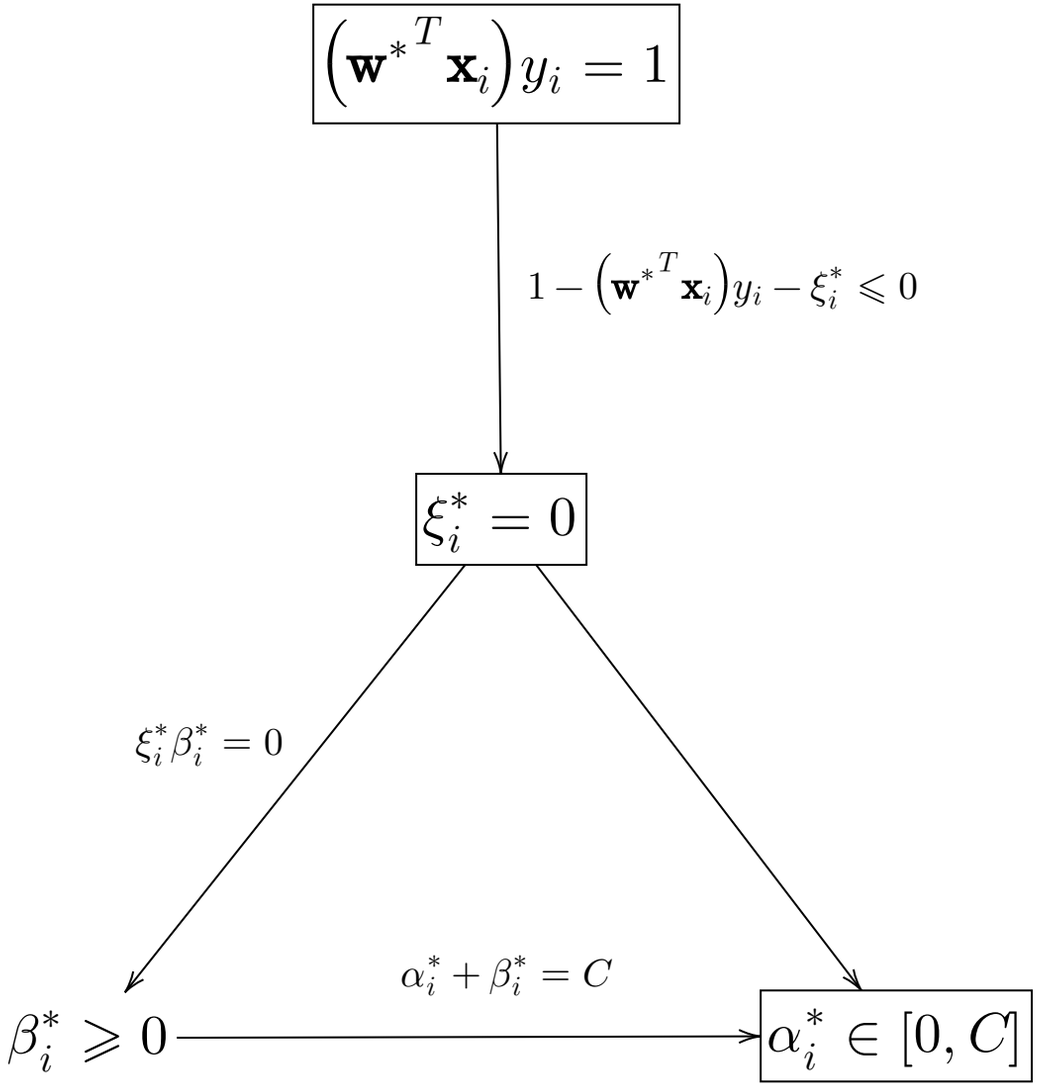
$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$





Soft-Margin, Complementary Slackness, Primal View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

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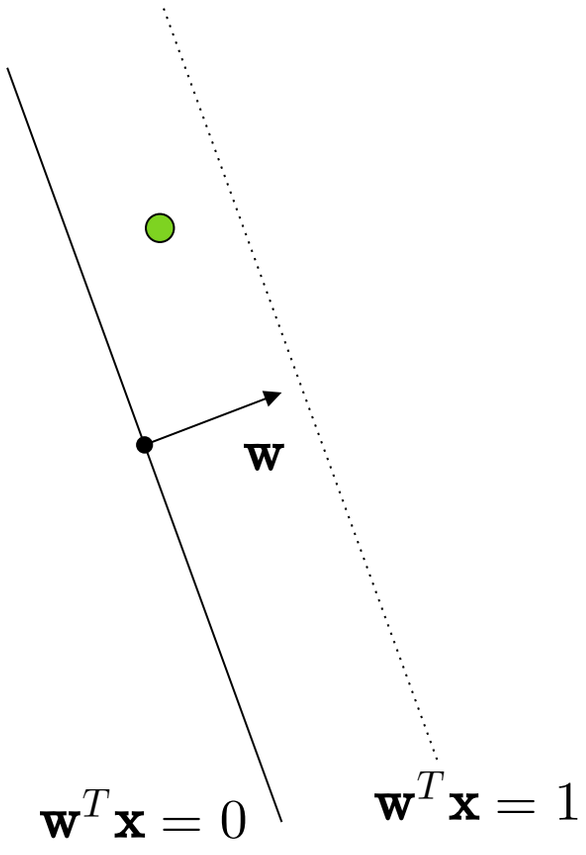

Soft-Margin, Complementary Slackness, Primal View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

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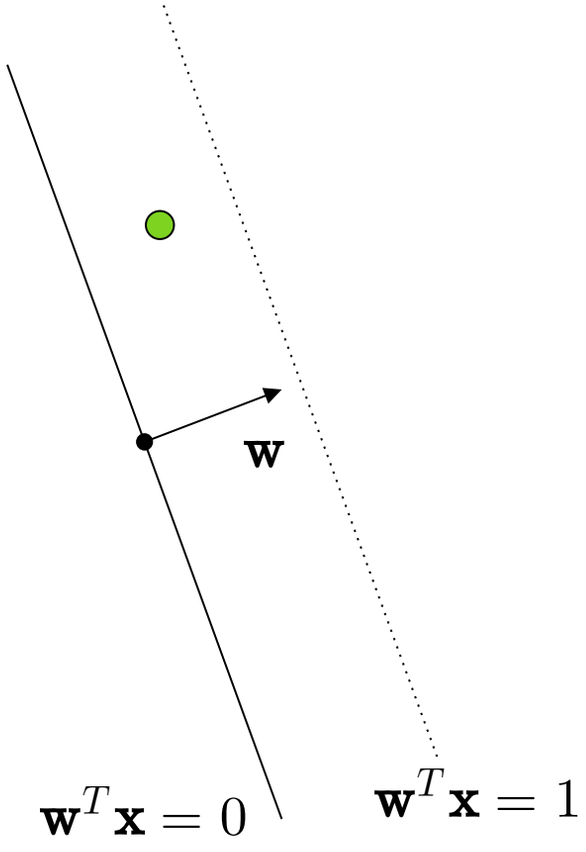


$$\boxed{\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i < 1}$$

Soft-Margin, Complementary Slackness, Primal View

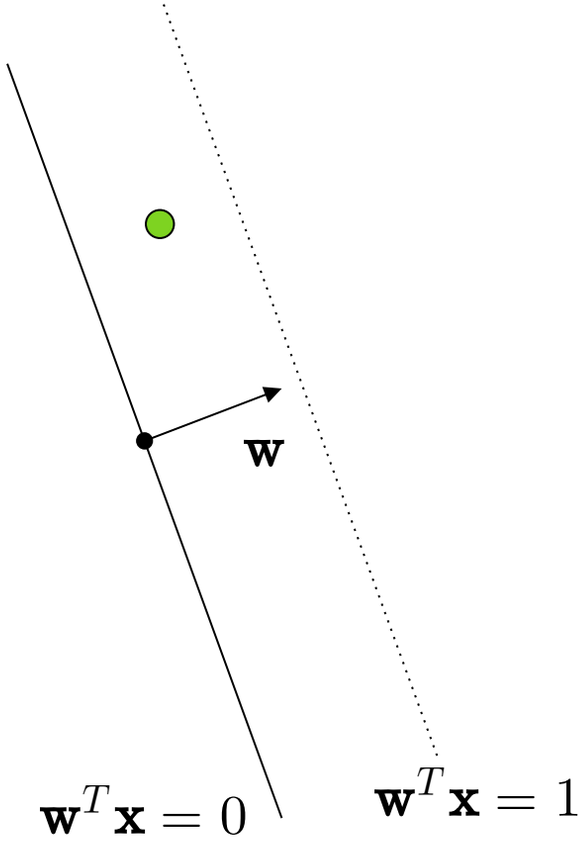
$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$



$$\boxed{\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i < 1}$$

$$1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^* \leqslant 0$$

$$\boxed{\xi_i^* > 0}$$

Soft-Margin, Complementary Slackness, Primal View

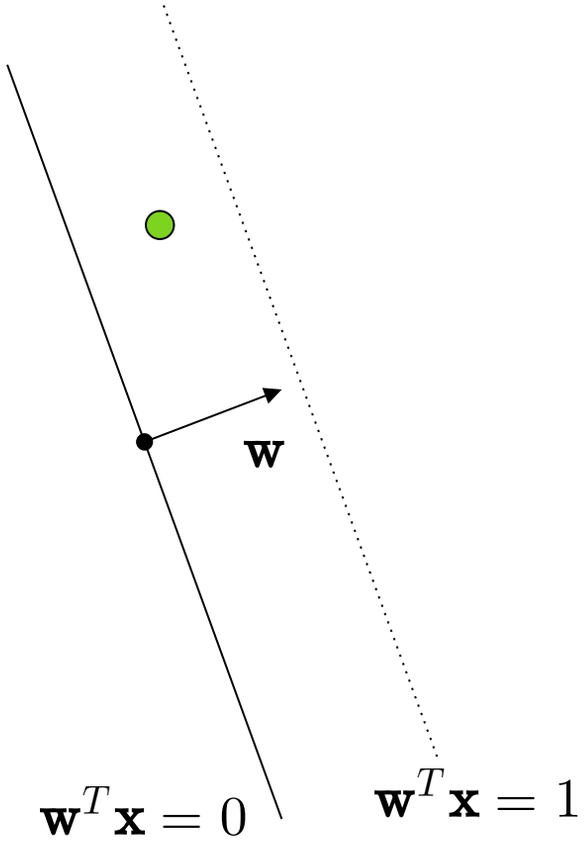
$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$



$$\boxed{\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i < 1}$$

$$1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^* \leqslant 0$$

$$\boxed{\xi_i^* > 0}$$

$$\xi_i^* \beta_i^* = 0$$

$$\beta_i^* = 0$$

Soft-Margin, Complementary Slackness, Primal View

$$\left(\mathbf{w}^{*T}\mathbf{x}_i\right)y_i + \xi_i^* \geqslant 1$$

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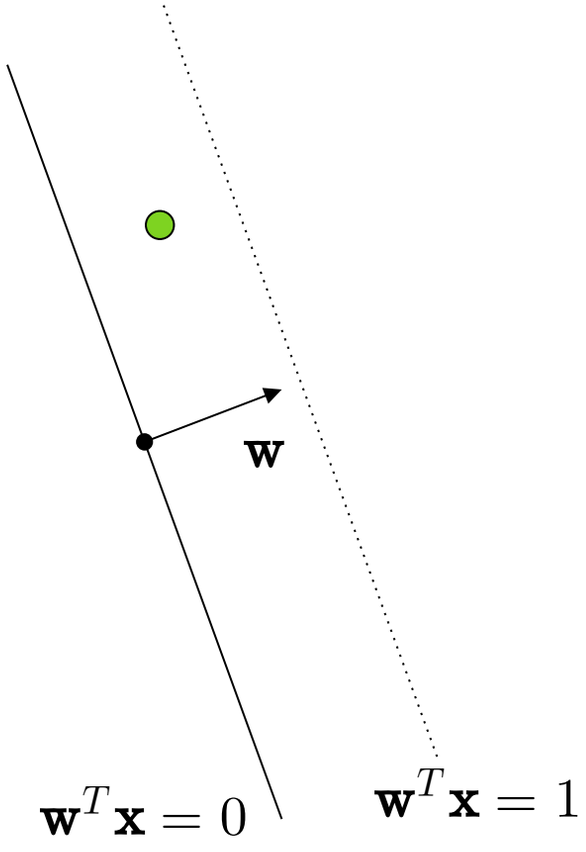

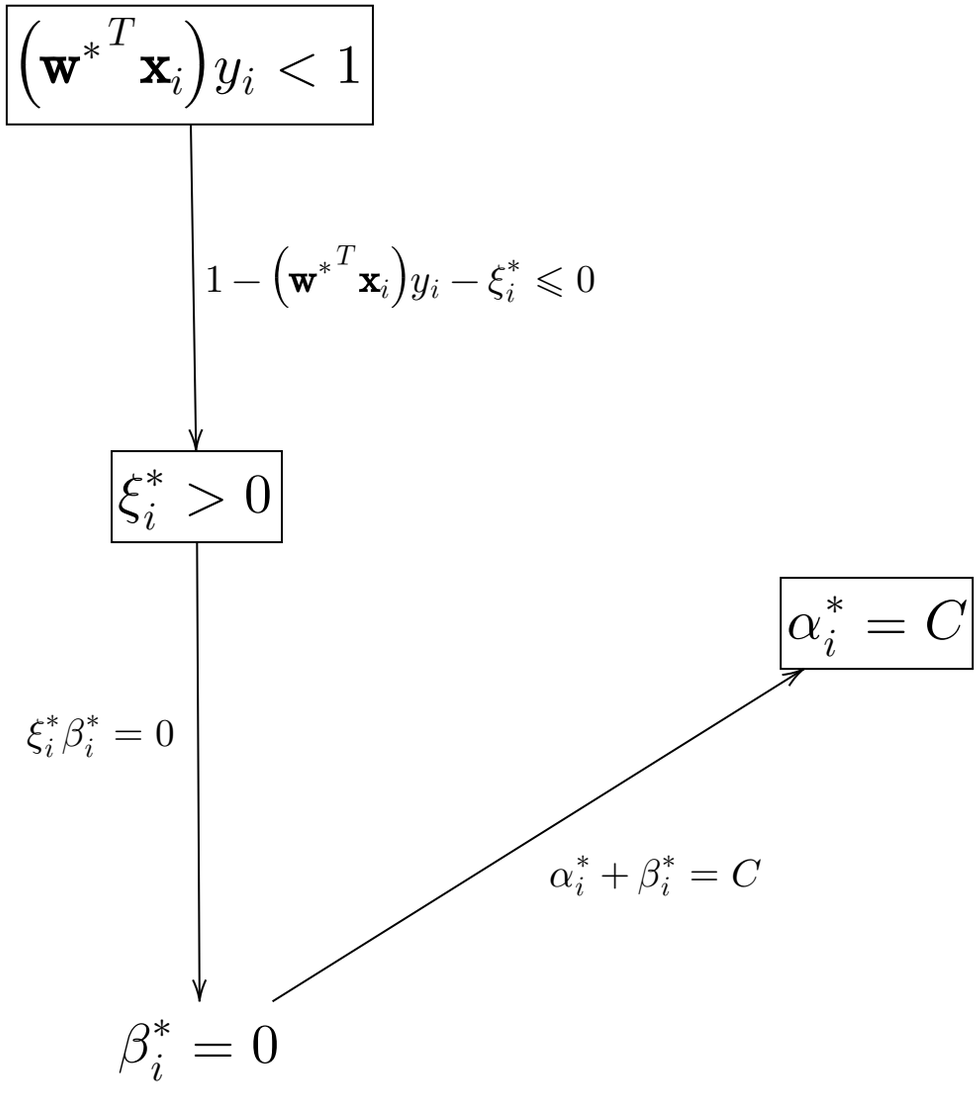

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$

$$\boxed{\alpha_i^* = 0}$$

Soft-Margin, Complementary Slackness, Dual View

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$$\boxed{\alpha_i^* = 0}$$

$$\xrightarrow{\alpha_i^* + \beta_i^* = C}$$

$$\beta_i^* = C$$

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

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$$\xi_i^* \beta_i^* = 0$$

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$$\boxed{\alpha_i^* = 0}$$

$$\xrightarrow{\alpha_i^* + \beta_i^* = C}$$

$$\beta_i^* = C$$

$$\xrightarrow{\xi_i^* \beta_i^* = 0}$$

$$\boxed{\xi_i^* = 0}$$

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

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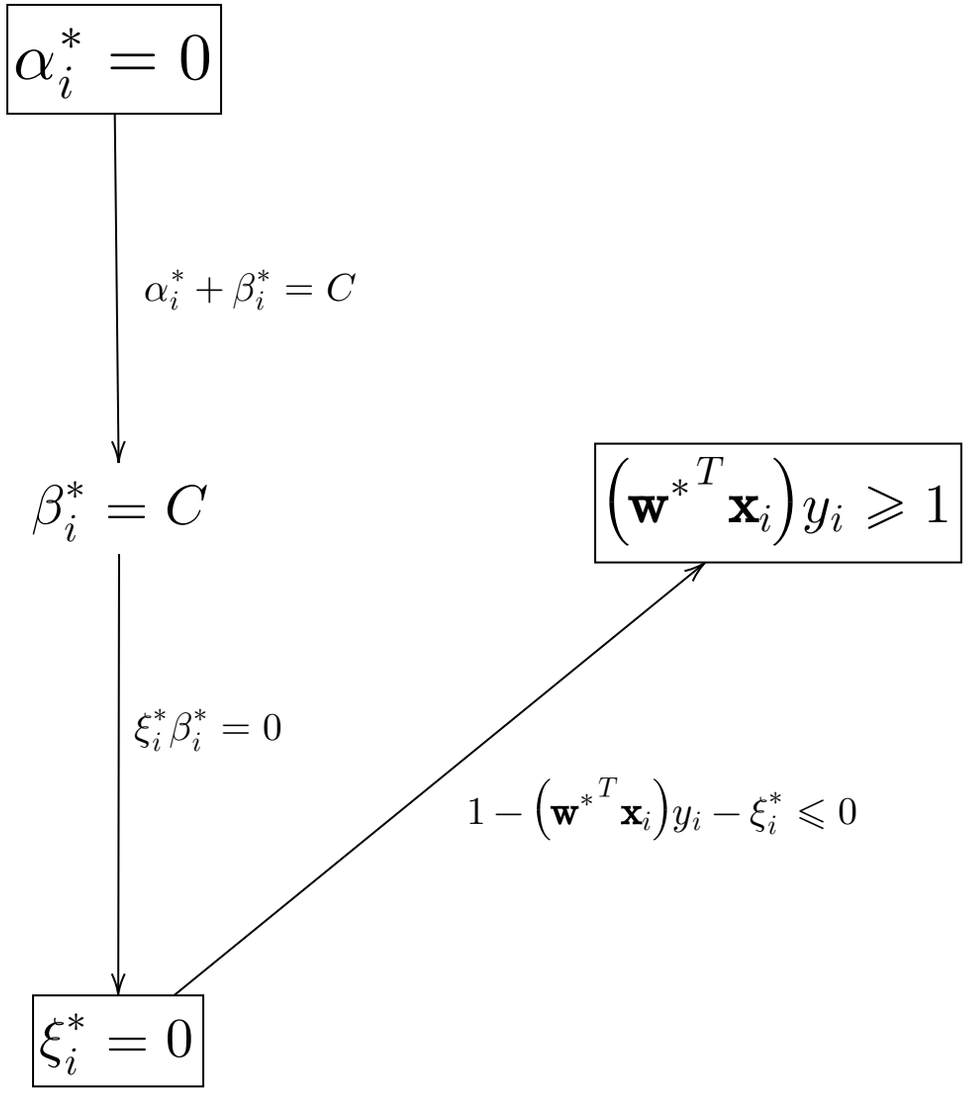

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

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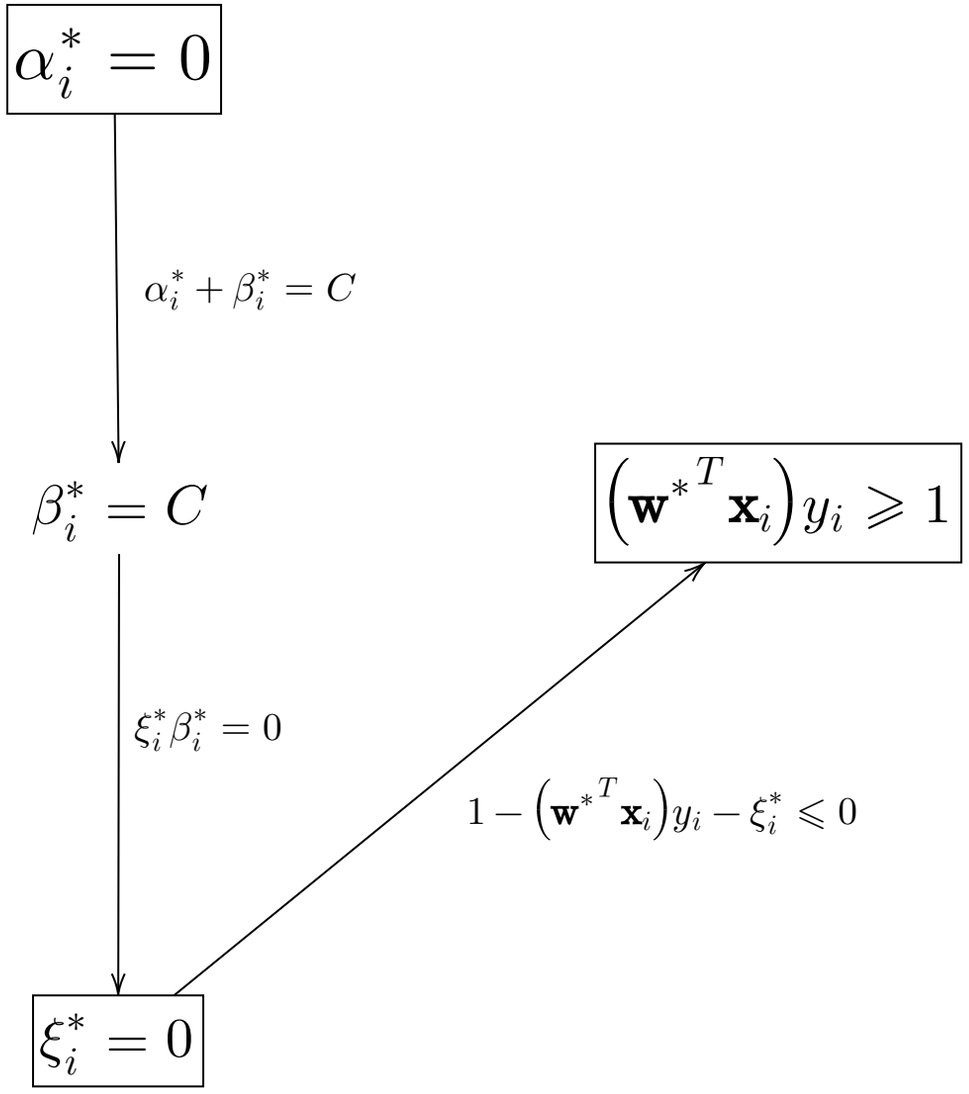

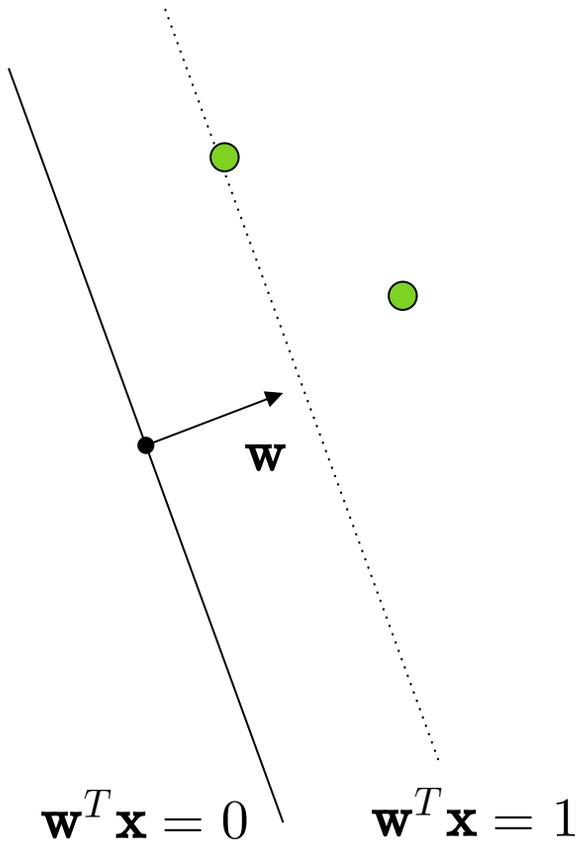

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$

$$\boxed{\alpha_i^* \in (0, C)}$$

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

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$$\boxed{\alpha_i^* \in (0, C)}$$

$$\xrightarrow{\alpha_i^* + \beta_i^* = C}$$

$$\beta_i^* \in (0, C)$$

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

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$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$

$$\boxed{\alpha_i^* \in (0, C)}$$

$$\Big\downarrow \; \alpha_i^* + \beta_i^* = C$$

$$\beta_i^* \in (0, C)$$

$$\Big\downarrow \; \xi_i^* \beta_i^* = 0$$

$$\boxed{\xi_i^* = 0}$$

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

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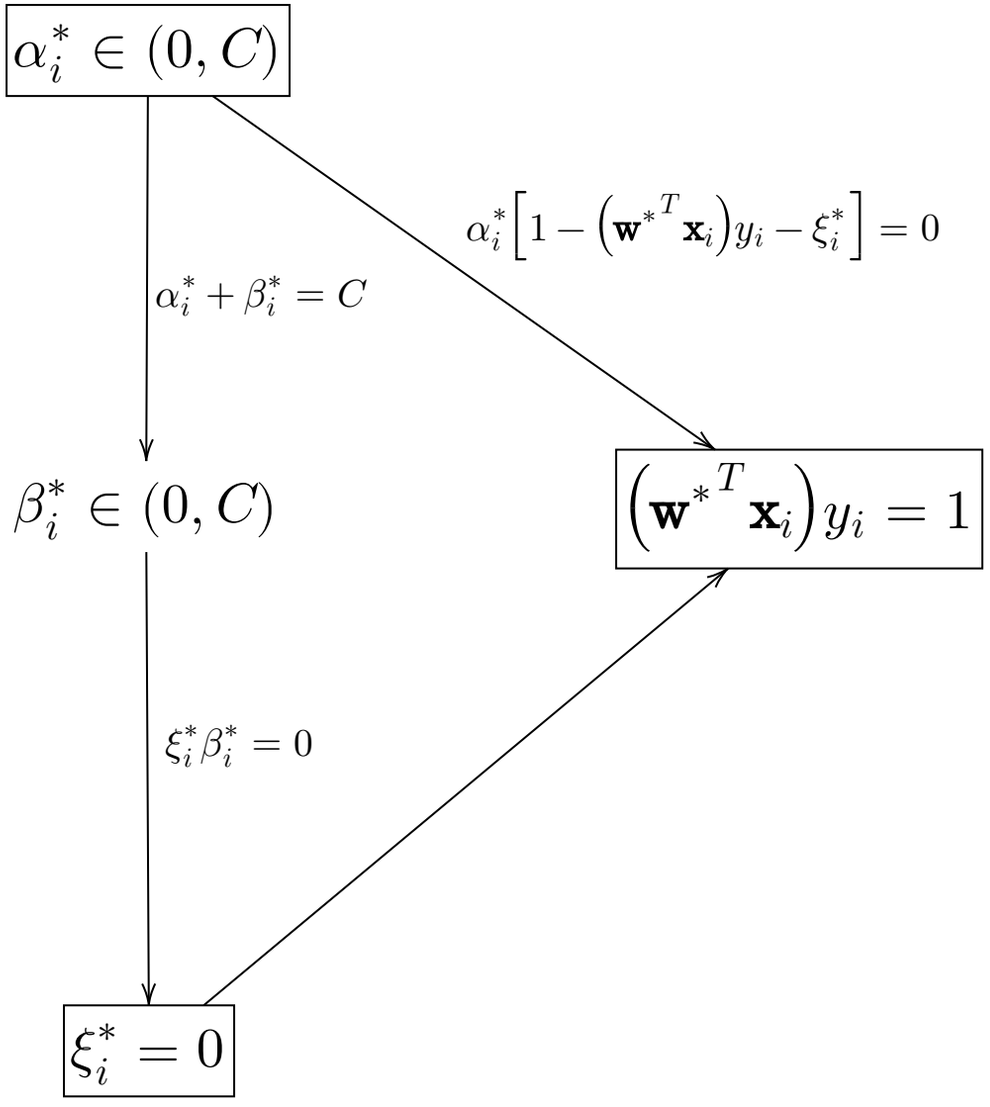

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

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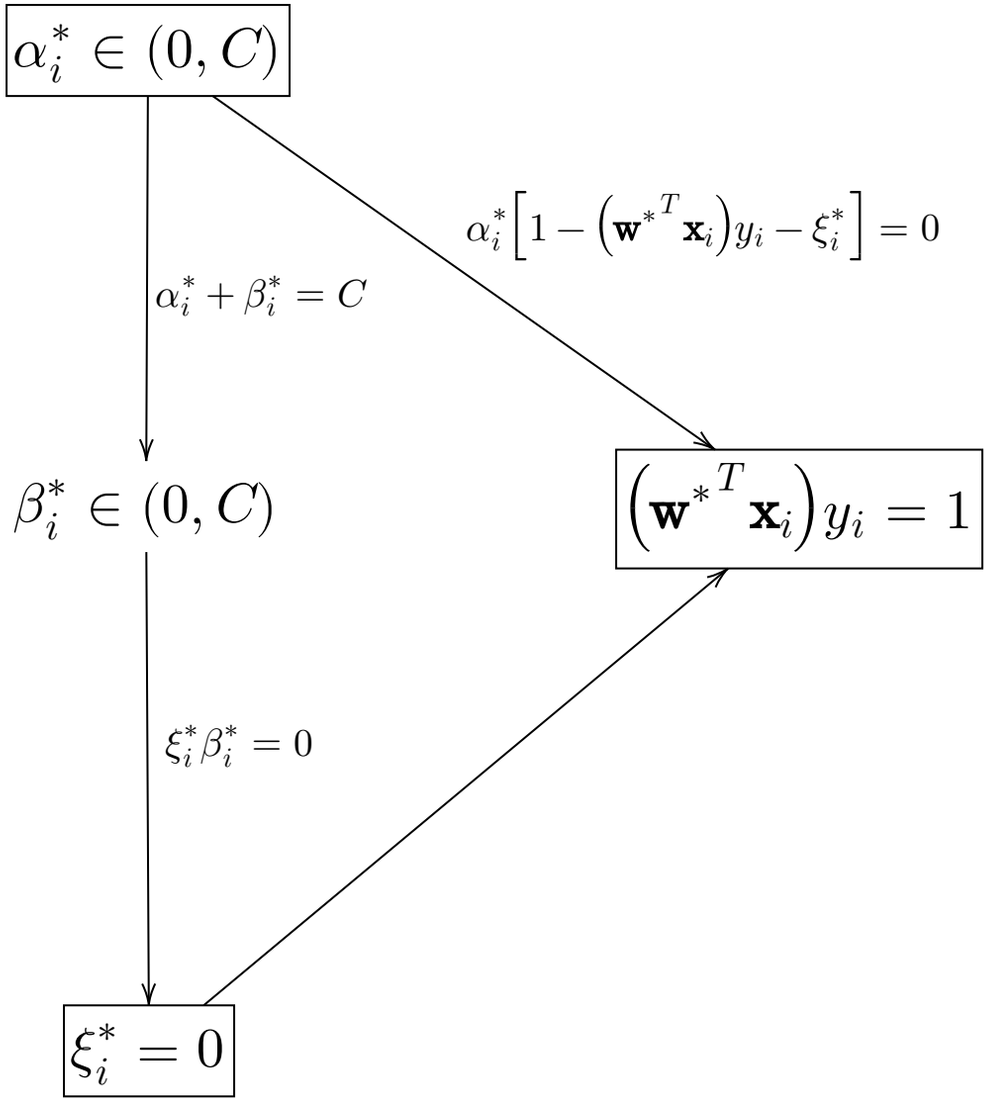

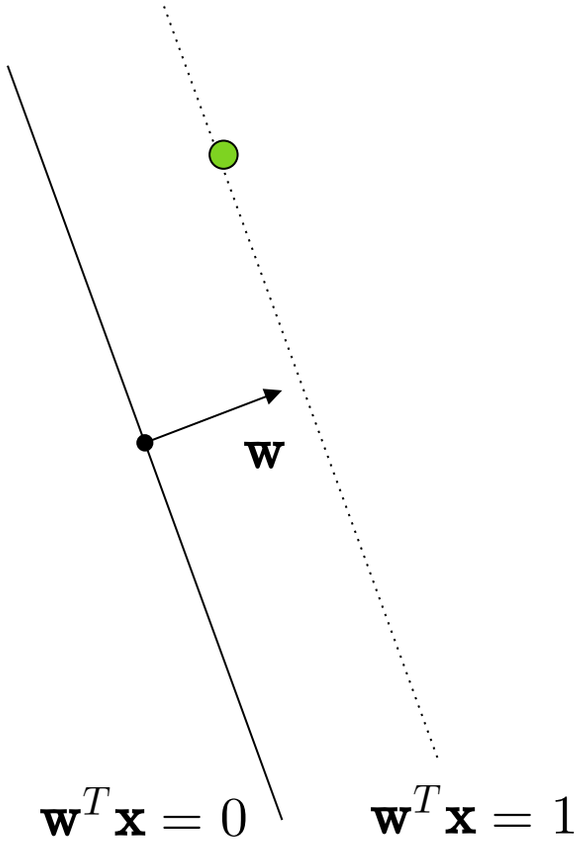

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

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$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$

$$\boxed{\alpha_i^* = C}$$

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$

$$\boxed{\alpha_i^* = C} \xrightarrow{\alpha_i^* + \beta_i^* = C} \beta_i^* = 0$$

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

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$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$

$$\boxed{\alpha_i^* = C}$$

$$\xrightarrow{\alpha_i^* + \beta_i^* = C}$$

$$\beta_i^* = 0$$

$$\xrightarrow{\xi_i^* \beta_i^* = 0}$$

$$\boxed{\xi_i^* \geqslant 0}$$

Soft-Margin, Complementary Slackness, Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

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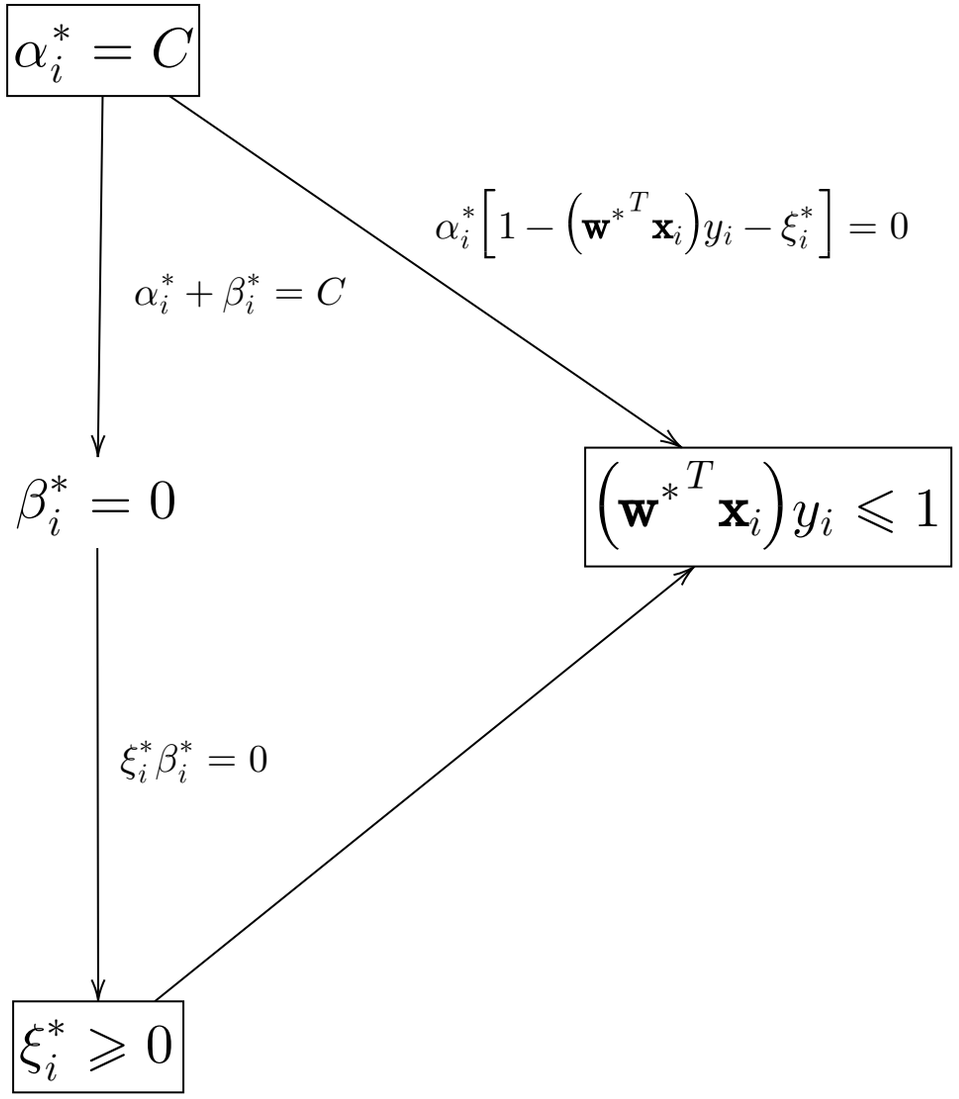

Soft-Margin, Complementary Slackness, Dual View

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$$\alpha_i^* + \beta_i^* = C$$

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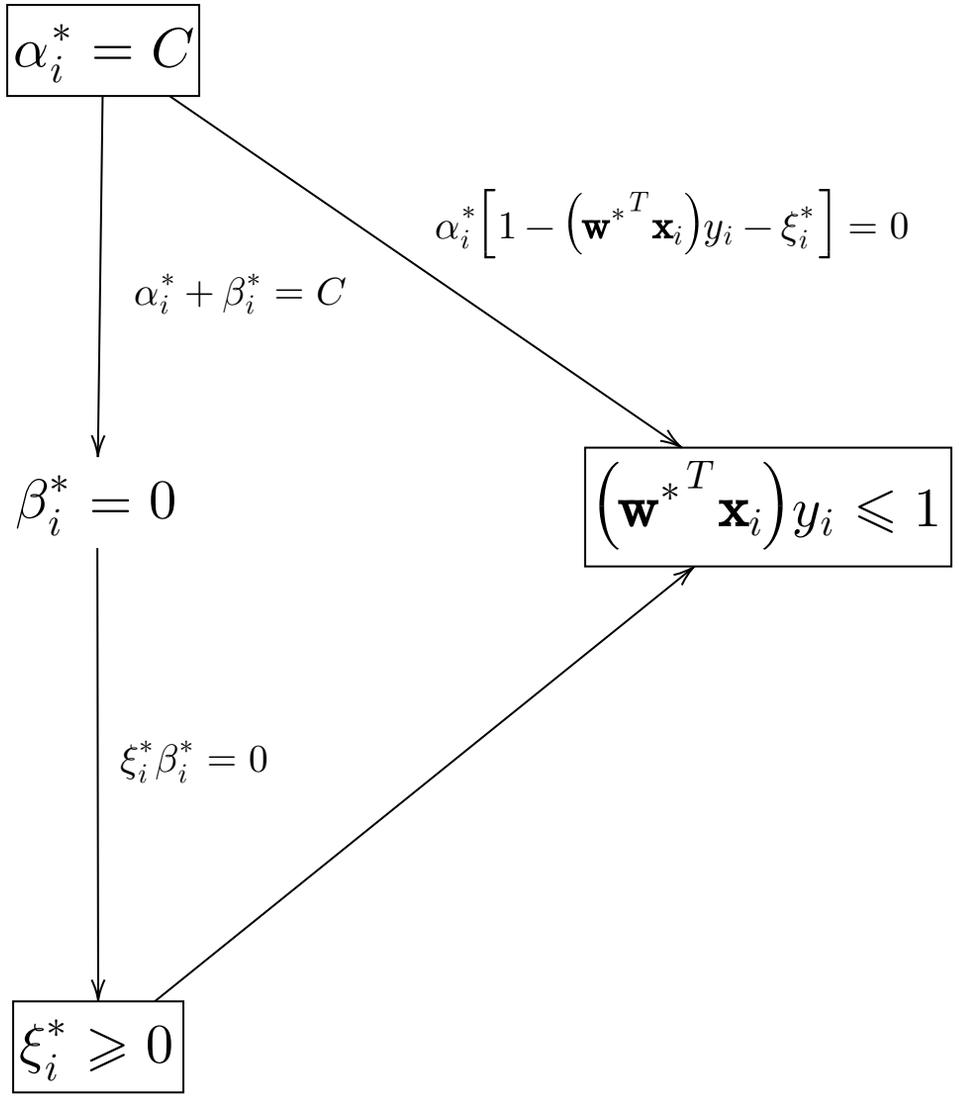

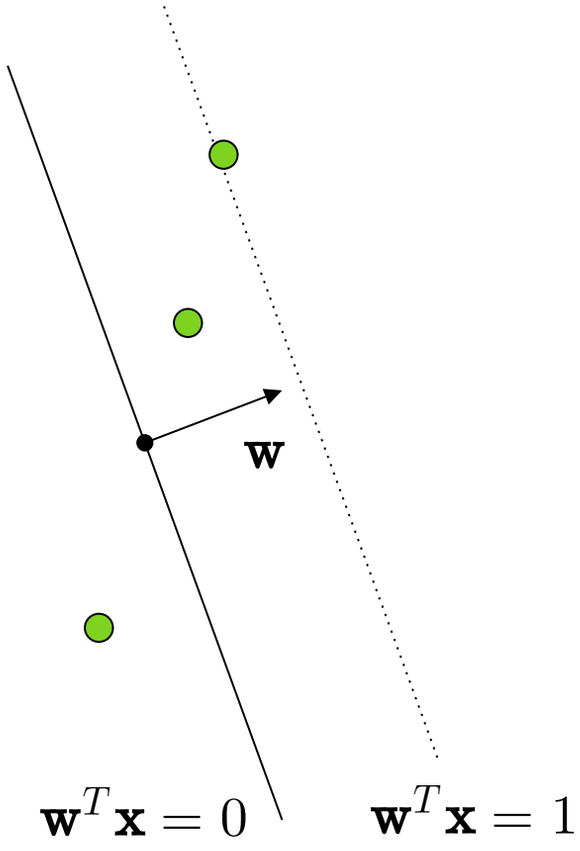

Soft-Margin, Complementary Slackness, Primal and Dual View

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$$

$$\xi_i^* \geqslant 0 \qquad \alpha_i^* \geqslant 0 \qquad \beta_i^* \geqslant 0$$

$$\alpha_i^* + \beta_i^* = C$$

$$\xi_i^* \beta_i^* = 0$$

$$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$$

Primal

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i > 1 \Longrightarrow \alpha_i^* = 0$$

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i = 1 \Longrightarrow \alpha_i^* \in [0, C]$$

$$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i < 1 \Longrightarrow \alpha_i^* = C$$

Soft-Margin, Complementary Slackness, Primal and Dual View

	Primal	Dual
$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i + \xi_i^* \geqslant 1$		
$\xi_i^* \geqslant 0 \quad \alpha_i^* \geqslant 0 \quad \beta_i^* \geqslant 0$	$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i > 1 \Longrightarrow \alpha_i^* = 0$	$\alpha_i^* = 0 \Longrightarrow \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i \geqslant 1$
$\alpha_i^* + \beta_i^* = C$		
$\xi_i^* \beta_i^* = 0$	$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i = 1 \Longrightarrow \alpha_i^* \in [0, C]$	$\alpha_i^* \in (0, C) \Longrightarrow \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i = 1$
$\alpha_i^* \left[1 - \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i - \xi_i^*\right] = 0$	$\left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i < 1 \Longrightarrow \alpha_i^* = C$	$\alpha_i^* = C \Longrightarrow \left({\mathbf{w}^*}^T \mathbf{x}_i\right) y_i \leqslant 1$

Miscellaneous

Terminology used

(1) Hard-margin, Linear-SVM
(2) Hard-margin, Kernel-SVM
(3) Soft-margin, Linear-SVM
(4) Soft-margin, Kernel-SVM

Additional points

- Discriminative model
- Weight vector is a sparse linear combination of data-points
- The dual is a quadratic programming problem